To my family, and to everyone who discussed a factor graph with me at improbable hours across improbable time zones.

Contents

Chapter 1

Introduction

1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). Their central object is the *score function* — the gradient of the log-density of progressively noised data,

$$S(x,t) \;=\; \nabla_x \log p_t(x), \tag{1.1}$$

which is all a model needs to know in order to reverse a noising process and turn pure noise back into data (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of what it has to learn remains poorly understood.

A complementary research programme, initiated in the statistical-physics community, replaces the black box with *exactly solvable models*: data distributions simple enough that the score can be computed in closed form, yet rich enough to exhibit the phenomena observed in real systems — speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024). Exact scores act as a microscope: every regime, transition time, and failure mode can be traced back to explicit formulas.

This thesis extends the exactly solvable programme in a direction that is natural for data with *temporal or sequential structure*: the data are not a cloud of independent points but a *trajectory* $a_0, a_1, \ldots, a_{K-1}$ of a Markov process — the prototype of a video, a time series, or any dynamic object. Each frame of the trajectory is corrupted independently by an Ornstein–Uhlenbeck (OU) channel, and the object of study is the *joint* score of the whole noisy sequence. Two classical bodies of theory then meet the modern one. First, the joint score of a Markov chain turns out to be governed by the algebra of structured matrices — Toeplitz covariances, tridiagonal precisions, and their spectral properties. Second, computing the score is an exercise in Bayesian inference on a tree-shaped factor graph, the home turf of *belief propagation* (BP) and its Gaussian projection, *approximate message passing* (AMP) (Pearl, 1988; Mézard and Montanari, 2009; Donoho et al., 2009). The thesis develops both routes in full detail, proves where they coincide, and measures precisely what is lost when they are forced apart.

1.2 Positioning and gap

The literature this work builds on can be organised in three strands.

Statistical physics of learning machines. Machine learning and statistical mechanics share a genealogy that runs from spin glasses through the Hopfield network and the Boltzmann machine to today's generative models, a lineage recognised by the 2024 Nobel Prize in Physics (Hopfield, 1982; Ackley et al., 1985; The Royal Swedish Academy of Sciences, 2024). Its modern branch produced closed-form analyses of diffusion generation for Gaussian mixtures, random energy models, and high-dimensional asymptotics (Biroli and Mézard, 2023; Biroli et al., 2024; Ghio et al., 2024), and, on the training side, the demonstration that energy-based models learn through a cascade of phase transitions (Bachtis et al., 2024). These works treat the data as unstructured collections of samples; the time axis they study is the *diffusion* (or training) time. The internal time of a *sequence* — the fact that a video frame is correlated with its neighbours through a dynamical rule — is absent from these analyses.

Inference on chains. Exact posterior inference on Gauss–Markov chains is classical: the Kalman filter and smoother (Kalman, 1960; Rauch et al., 1965), hidden Markov models (Rabiner, 1989), and their unifying formulation as sum–product message passing on factor graphs (Kschischang et al., 2001; Mézard and Montanari, 2009). What this literature does not ask is the question that matters for generative modelling: *how does the exact posterior — and therefore the exact score — deform as a function of the diffusion time*, and what does that deformation imply for the architecture of a network that must learn it?

Message passing with continuous variables. BP is exact on trees for discrete variables, where messages are finite vectors of probabilities. For continuous variables the messages become probability *densities* — infinite-dimensional objects — and practical algorithms must close them within a parametric family, almost always the Gaussian one; this is the step that turns BP into AMP, known in physics as the Thouless–Anderson–Palmer (TAP) approach (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016). The classical justification of the Gaussian closure is a central-limit argument valid when every node receives many messages. On a chain each node receives exactly two: the central-limit argument fails by construction, and the quality of the Gaussian closure on chain-structured diffusion posteriors was, to the best of our knowledge, unquantified.

The gap, in one sentence: *there is no end-to-end, exactly solvable account of the joint diffusion score of a structured sequence — its matrix structure, its message-passing computation, and the price of the Gaussian closure when the sequence is not Gaussian.*

1.3 Research questions

The thesis addresses four questions.

RQ1. What is the exact joint score of a Markov chain corrupted by an OU channel, and how does its structure — concretely, the posterior precision matrix — evolve along the diffusion time, from the sparse Markov regime at $t \to 0$ to the isotropic regime at $t \to \infty$?

RQ2. Can the joint score be computed *locally*, by message passing on the chain's factor graph, rather than globally by matrix inversion? In the Gaussian case, how does belief propagation relate to the classical Kalman smoother, and does it reproduce the closed-form score exactly?

RQ3. When the messages are forced into a Gaussian family — the AMP/TAP closure — what survives and what breaks? On the Gaussian chain, is the closure exact; and on a non-Gaussian (Laplace-innovation) chain, how large is the closure error as a function of the diffusion time?

RQ4. What do the exact results imply for *learned* scores: when is a local (banded, convolution-like) score architecture sufficient, and what is the exact law governing the error of local truncations?

1.4 Contributions

The main contributions, each developed in a dedicated chapter, are the following.

1. **A complete closed-form solution of the Gaussian benchmark** (Chapter 5). For the stationary AR(1) chain under OU corruption we derive the joint score, diagonalise it in closed form, and characterise the full lifecycle of the posterior precision matrix: a corrected *band-fill law* for the small-t growth of its off-diagonals, and exact bulk formulas for its inverse — variance, geometric decay rate $q(\alpha, t)$ of posterior correlations, and their limits. The band-fill law corrects a transcription error in an earlier working note; the correction is validated numerically.

2. **An equation-by-equation message-passing construction of the score** (Chapter 7). We build the factor graph of the noisy chain, specialise the sum–product

equations to it, prove that Gaussian messages are closed under the updates (two elementary facts suffice), and show that the resulting algorithm — Gaussian BP, algebraically identical to a Kalman smoother — reproduces the matrix score to machine precision at cost $O(K)$ instead of $O(K^3)$.

3. **An exact phase boundary for the AMP closure** (Appendix C). On the chain, BP and AMP produce the *same score* but *different posterior variances.* We show the difference is a single factor of two in a closure discriminant ($J_d^2 - 4\beta^2$ for BP against $J_d^2 - 8\beta^2$ for AMP), prove that the BP fixed point exists for every coupling and diffusion time, and derive the exact AMP breakdown time $t_c(\alpha)$ together with the critical coupling $\alpha_c = \sqrt{2} - 1$ below which AMP never breaks down. These results are developed in an appendix because they branch off the thesis's main line; they are complete and fully audited.

4. **A quantitative audit of the Gaussian closure beyond Gaussianity** (Chapters 6 and 7). For Laplace-innovation chains — where exact inference is still available through deterministic grid quadrature on the same factor graph — we measure the score error of Gaussian-projected BP: it peaks around 20% (relative L^2) at small diffusion time and decays below 1% as the noisy marginals Gaussianise, while the score *direction* remains accurate throughout. We also show that at small t *truncating the inference* (local BP sweeps) is markedly more accurate than *truncating the score matrix* (banding), quantifying the advantage of local message passing over local linear algebra.

5. **A reproducible computational companion.** All results are backed by open, dependency-light Python code with deterministic numerical audits (72/72 and 55/55 independent checks, re-executed on the build machine); the code and the figures it generates are part of the thesis repository, documented in Appendix D.

1.5 Methodology

The methodology is deliberately old-fashioned: *no neural networks, no black boxes — closed forms first, numerics as audit.* Every claim is either proved by explicit algebra (the derivations are spelled out step by step, with the longer algebraic passages isolated in numbered *toolboxes*) or verified by deterministic numerical checks against brute-force ground truth, in the spirit of the exactly solvable programme of Biroli and Mézard (2023). Where an approximation is introduced (the Gaussian closure of BP messages, local truncations), its error is measured against an exact reference computed by an independent method (matrix algebra or dense grid quadrature). The empirical setting, the audit protocol, and the software packages are described alongside the results and in Appendix D.

1.6 Structure of the thesis

The thesis is organised in two movements: a literature-review movement (Chapters 2–4) that tells the story this work belongs to and builds every tool it uses, and a research movement (Chapters 5–7) that contains the original contributions. Throughout, the rule is that theory is developed where it is needed, at the depth its use demands, with longer algebraic passages isolated in numbered *toolboxes* and full derivations of supporting results collected in the appendices.

Chapter 2 develops the statistical mechanics this thesis uses — phase space and Liouville's theorem, the Boltzmann–Gibbs distribution, free energy, the Ising model — and then follows the discipline into machine learning: spin glasses, the Hopfield network, the Boltzmann machine and its restricted variant, and the statistical mechanics of learning, up to the modern results on training-time phase transitions. Chapter 3 supplies the probabilistic dynamics: Markov chains, the AR(1) running example, sampling by dynamics, the Itô calculus, the Ornstein–Uhlenbeck corruption channel, the Fokker–Planck equation, and Anderson's time-reversal theorem. Chapter 4 reviews modern generative modelling in depth: energy-based models in LeCun's formulation, score matching, the DDPM construction, the score-SDE synthesis, flow models, the score as a posterior expectation, and the statistical-physics analyses of diffusion — including the cascade of phase transitions in energy-based-model training — that frame our research questions.

Chapter 5 solves the Gaussian chain exactly (RQ1), with the spectral form of the precision matrix as the organising object. Chapter 6 leaves Gaussianity in the most controlled way possible — Laplace innovations — and shows what the joint score looks like when it is no longer linear (first half of RQ3). Chapter 7 develops belief propagation and the Kalman filter in full, proves the Gaussian closure, and measures the closure error on non-Gaussian chains (RQ2, RQ3) together with the locality laws that answer RQ4. Chapter 8 summarises the findings per research question, discusses limitations, and lays out the research programme that follows. Four appendices collect Gaussian identities, long derivations, the AMP/TAP analysis with its exact phase boundary, and the reproducibility protocol.

Chapter 2

Statistical Mechanics and the Origins of Learning Machines

Statistical mechanics is the discipline that explains how probabilistic laws emerge from deterministic dynamics. It is also the field in which the neural network was invented. This chapter serves both facts, and its two halves have distinct jobs. The first half builds the small set of physical tools that the rest of the thesis actually uses — the Boltzmann distribution, the partition function and free energy, and the Ising energy — with the derivations carried out in full, because Chapters 3 and 4 lean on each of them: the Boltzmann distribution is the stationary law of Langevin dynamics, the free energy is the variational principle behind the diffusion-model training objective, and *sampling by simulating a physical process* is statistical mechanics turned into an algorithm. The second half is the part of the literature review that most histories of machine learning skip: how spin glasses became the Hopfield network, how the Hopfield network became the Boltzmann machine, and how the statistical mechanics of these systems grew into the modern theory of learning — the tradition, recognised by the 2024 Nobel Prize in Physics, in which this thesis works. The physical presentation follows classical references (Goldstein et al., 2002; Landau and Lifshitz, 1980).

2.1 Hamiltonian mechanics and phase space

Consider a mechanical system with n degrees of freedom, described by generalised coordinates $q = (q_1, \ldots, q_n)$ and conjugate momenta $p = (p_1, \ldots, p_n)$. Its dynamics is generated by a single scalar function, the *Hamiltonian* $H(q, p)$, which for the systems considered here has the mechanical form

$$H(q,p) \;=\; \underbrace{\sum_{i=1}^{n} \frac{p_i^2}{2m_i}}_{\text{kinetic}} \;+\; \underbrace{U(q_1,\ldots,q_n)}_{\text{potential}}, \tag{2.1}$$

and equals the total energy of the system. The state of the system at any instant is the pair (q, p), a point in the $2n$-dimensional *phase space* $\Gamma \subseteq \mathbb{R}^{2n}$. The evolution is given by *Hamilton's equations*,

$$\dot{q}_i \;=\; \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i \;=\; -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, n. \tag{2.2}$$

Two structural properties of (2.2) are the pillars on which statistical mechanics rests.

Energy conservation. Along any trajectory,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i}\,\dot{q}_i + \frac{\partial H}{\partial p_i}\,\dot{p}_i \right) = \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i} \right) = 0. \tag{2.3}$$

The motion is therefore confined to the *energy shell* $\Gamma_E = \{(q,p) : H(q,p) = E\}$, a $(2n-1)$-dimensional hypersurface.

Incompressibility of the phase flow. The phase-space velocity field

$$v(q,p) \;=\; (\dot{q}, \dot{p}) \;=\; \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q} \right) \tag{2.4}$$

has identically vanishing divergence:

$$\nabla \cdot v \;=\; \sum_{i=1}^{n} \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) \;=\; \sum_{i=1}^{n} \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) \;=\; 0. \tag{2.5}$$

Hamiltonian flow behaves like an incompressible fluid in phase space: it can stretch and fold a region, but never change its volume. This innocuous-looking identity is the seed of the entire probabilistic edifice, as we now show.

2.2 Liouville's theorem

A single trajectory is useless for macroscopic physics: for a gas, $n \sim 10^{23}$ and neither the initial condition nor the solution of (2.2) is accessible. The move that founds statistical mechanics is to describe our *ignorance* of the microstate by a probability density $\rho(q, p, t)$ on phase space: $\rho(q, p, t)\,\mathrm{d}q\,\mathrm{d}p$ is the probability that the system is in the infinitesimal cell around (q, p) at time t. Liouville's theorem states how this density evolves.

Theorem 2.1 (Liouville)**.** *Under the Hamiltonian flow* (2.2)*, the phase-space density obeys*

$$\frac{\partial \rho}{\partial t} \;=\; -\{\rho, H\} \;=\; \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i} \frac{\partial \rho}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial \rho}{\partial q_i} \right), \tag{2.6}$$

where $\{\cdot, \cdot\}$ *is the Poisson bracket. Equivalently, the density is constant along trajectories:* $\mathrm{d}\rho/\mathrm{d}t = 0$.

Toolbox 2.1 — Proof of Liouville's theorem, step by step

The density is locally conserved — probability mass is neither created nor destroyed, only transported by the flow — so it satisfies the continuity equation familiar from fluid dynamics:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\, v) = 0, \qquad v = (\dot{q}, \dot{p}).$$

Expand the divergence of the product using $\nabla \cdot (\rho v) = v \cdot \nabla\rho + \rho\,(\nabla \cdot v)$:

$$\frac{\partial\rho}{\partial t} + \underbrace{\sum_{i=1}^{n}\left(\dot{q}_i\frac{\partial\rho}{\partial q_i} + \dot{p}_i\frac{\partial\rho}{\partial p_i}\right)}_{v\cdot\nabla\rho} + \rho\underbrace{\sum_{i=1}^{n}\left(\frac{\partial\dot{q}_i}{\partial q_i} + \frac{\partial\dot{p}_i}{\partial p_i}\right)}_{\nabla\cdot v} = 0.$$

By the incompressibility identity (2.5) the last term vanishes. Substituting Hamilton's equations $\dot{q}_i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q_i$ into the middle term:

$$\frac{\partial\rho}{\partial t} + \sum_{i=1}^{n}\left(\frac{\partial H}{\partial p_i}\frac{\partial\rho}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial\rho}{\partial p_i}\right) = 0 \quad\Longleftrightarrow\quad \frac{\partial\rho}{\partial t} = -\{\rho, H\}.$$

Finally, the total (convective) derivative along a trajectory is

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial\rho}{\partial t} + v\cdot\nabla\rho = \frac{\partial\rho}{\partial t} + \{\rho, H\} = 0,$$

which is the second form of the statement: an observer riding a phase-space trajectory sees a constant local density. □

Liouville's theorem has an immediate and profound consequence: *any density that is a function of conserved quantities only is stationary.* If $\rho(q,p) = f\big(H(q,p)\big)$ for some function f, then $\{\rho, H\} = f'(H)\,\{H, H\} = 0$, so $\partial\rho/\partial t = 0$. Equilibrium statistical mechanics is the study of which f nature chooses.

2.3 From dynamics to ensembles

An *ensemble* is a probability distribution over microstates that is stationary under the dynamics and compatible with the macroscopic constraints we impose (fixed energy, fixed temperature, . . .). Two ensembles matter for this thesis.

2.3.1 The microcanonical ensemble

For a perfectly isolated system, the energy is fixed at E and Liouville's theorem suggests the simplest choice compatible with it: a uniform density on the energy shell,

$$\rho_{\mathrm{mc}}(q,p) \;=\; \frac{1}{\Omega(E)}\,\delta\Big(H(q,p)-E\Big), \qquad \Omega(E)=\int_\Gamma \delta\Big(H(q,p)-E\Big)\,\mathrm{d}q\,\mathrm{d}p, \tag{2.7}$$

where $\Omega(E)$ counts the microstates at energy E. The postulate that all accessible microstates are equally likely — the *equal a priori probability postulate* — is motivated (not proved) by ergodicity: for ergodic dynamics, time averages along a single trajectory converge to averages over the shell, so the uniform measure is the one a long experiment effectively samples. Boltzmann's entropy attaches a number to this count,

$$S(E) \;=\; k_B \log \Omega(E), \tag{2.8}$$

with k_B the Boltzmann constant. Entropy is the logarithm of the number of microscopic ways a macroscopic state can be realised.

2.3.2 The canonical ensemble and the Boltzmann distribution

Isolated systems are an idealisation; the physically ubiquitous situation is a system exchanging energy with a much larger environment (a *heat bath*) at temperature T. The stationary distribution of the system alone is then no longer uniform: it is the *Boltzmann–Gibbs distribution*,

$$\boxed{\rho_\beta(x) \;=\; \frac{1}{Z(\beta)}\,e^{-\beta H(x)}, \qquad Z(\beta)=\int e^{-\beta H(x)}\,\mathrm{d}x, \qquad \beta=\frac{1}{k_B T},} \tag{2.9}$$

where $x=(q,p)$ collects the microstate and $Z(\beta)$, the *partition function*, normalises the density. Equation (2.9) can be derived in two independent ways, and both derivations are instructive enough to be carried out in full: the first (physical) traces the exponential to the entropy of the bath; the second (information-theoretic) characterises it as the leastcommittal distribution compatible with a fixed mean energy, a perspective due to Jaynes (1957) that returns, almost verbatim, in modern machine learning.

Toolbox 2.2 — The Boltzmann distribution from the heat bath

Let the total system = (system S) $\cup$ (bath B) be isolated with total energy E_{tot}, and let the coupling be weak, so $H_{\text{tot}} = H_S + H_B$. Apply the microcanonical postulate to the total system: the probability that S is in a *specific* microstate x with energy $H_S(x) = \epsilon$ is proportional to the number of bath microstates carrying the remaining energy,

$$\mathbb{P}(x) \;\propto\; \Omega_B\Big(E_{\text{tot}} - \epsilon\Big) \;=\; \exp\Big[\tfrac{1}{k_B} S_B\Big(E_{\text{tot}} - \epsilon\Big)\Big]\,,$$

using the definition (2.8) of the bath entropy. Since the bath is huge, $\epsilon \ll E_{\text{tot}}$, expand S_B to first order:

$$S_B(E_{\text{tot}} - \epsilon) = S_B(E_{\text{tot}}) - \epsilon\,\frac{\partial S_B}{\partial E}\bigg|_{E_{\text{tot}}} + O\Big(\epsilon^2/E_{\text{tot}}\Big).$$

The derivative of the bath entropy with respect to its energy *defines* the bath temperature, $\partial S_B/\partial E = 1/T$. Substituting,

$$\mathbb{P}(x) \;\propto\; \underbrace{e^{S_B(E_{\text{tot}})/k_B}}_{\text{constant in } x} \cdot e^{-\epsilon/(k_B T)} \;\propto\; e^{-\beta H_S(x)}.$$

The quadratic remainder is suppressed by the bath's heat capacity and vanishes in the thermodynamic limit. Normalising yields (2.9). Note what happened: the exponential form is nothing but the first-order Taylor expansion of the bath's entropy — the Boltzmann factor is the price, in bath microstates, of lending energy ϵ to the system. □

Toolbox 2.3 — The Boltzmann distribution from maximum entropy

Forget physics; ask, with Jaynes (1957): among all densities ρ with a given mean energy $\mathbb{E}_\rho[H] = U$, which one assumes the least beyond the constraint? "Least" is measured by the Gibbs–Shannon entropy $S[\rho] = -k_B \int \rho \log \rho$. Maximise $S[\rho]$ subject to $\int \rho = 1$ and $\int \rho H = U$ with Lagrange multipliers λ_0, λ_1:

$$\mathcal{L}[\rho] = -k_B \int \rho \log \rho \, \mathrm{d}x \; - \; \lambda_0\Big(\int \rho \, \mathrm{d}x - 1\Big) \; - \; \lambda_1\Big(\int \rho H \, \mathrm{d}x - U\Big).$$

Take the functional derivative and set it to zero pointwise:

$$\frac{\delta \mathcal{L}}{\delta \rho(x)} = -k_B\Big(\log \rho(x) + 1\Big) - \lambda_0 - \lambda_1 H(x) = 0 \quad \Longrightarrow \quad \rho(x) = \exp\Big[-1 - \tfrac{\lambda_0}{k_B} - \tfrac{\lambda_1}{k_B} H(x)\Big].$$

The density is an exponential of the constrained quantity — this is completely general: constraining $\mathbb{E}[\phi_j(x)]$ for features ϕ_j produces $\rho \propto \exp[-\sum_j \beta_j \phi_j(x)]$, the exponential family. The multiplier λ_0 is fixed by normalisation and produces Z; renaming $\lambda_1/k_B = \beta$ and identifying $\beta = 1/(k_B T)$ through thermodynamics recovers (2.9) exactly. Second-order check: $\delta^2 \mathcal{L}/\delta \rho^2 = -k_B/\rho < 0$, so the stationary point is the unique maximum (the entropy is strictly concave). □

The second derivation deserves a remark that will echo through the thesis: the Boltzmann distribution is not a statement about molecules but about *inference under constraints*. Any time we posit a model of the form $p(x) \propto e^{-E(x)}$ — as we will for Markov random fields and energy-based models in Chapter 4 — we are doing statistical mechanics, whether the variables are spins, pixels, or the frames of a video.

2.4 The partition function and free energy

The partition function $Z(\beta)$ in (2.9) looks like a mere normaliser; it is in fact the central computational object of the theory — and the reason approximate inference exists as a field. Its logarithm defines the *Helmholtz free energy*,

$$F(\beta) \;=\; -\frac{1}{\beta} \log Z(\beta). \tag{2.10}$$

Toolbox 2.4 — Everything is a derivative of $\log Z$

Differentiate $\log Z$ with respect to β, carrying the derivative through the integral:

$$-\frac{\partial \log Z}{\partial \beta} = -\frac{1}{Z}\frac{\partial}{\partial \beta}\int e^{-\beta H} = \frac{\int H\, e^{-\beta H}}{\int e^{-\beta H}} = \mathbb{E}_{\rho_\beta}[H] \;=\; U,$$

the mean energy. Differentiate once more:

$$\frac{\partial^2 \log Z}{\partial \beta^2} = \frac{\partial}{\partial \beta}\left(-\frac{\int He^{-\beta H}}{Z}\right) = \frac{\int H^2 e^{-\beta H}}{Z} - \left(\frac{\int He^{-\beta H}}{Z}\right)^2$$
$$= \mathbb{E}[H^2] - \mathbb{E}[H]^2 \;=\; \mathrm{Var}(H) \;\geq\; 0.$$

So $\log Z$ is convex in β, its first derivative gives the mean energy, its second the energy fluctuations; the latter equals $k_B T^2 C_V$ with C_V the heat capacity — a *fluctuation–dissipation* relation: how a system responds to heating is determined by its spontaneous equilibrium fluctuations. Finally, rewriting the Gibbs–Shannon entropy of ρ_β:

$$S[\rho_\beta] = -k_B\mathbb{E}\Big[\log \rho_\beta\Big] = -k_B\mathbb{E}\Big[-\beta H - \log Z\Big] = \frac{U}{T} + k_B \log Z,$$

and multiplying by $-T$: $-TS = -U + F$, i.e.

$$F = U - TS.$$

The free energy is the exact trade-off between energy and entropy: at low T minimising F means minimising energy (ordered states win), at high T it means maximising entropy (disordered states win). Phase transitions are the non-analyticities of F where the winner changes. □

There is one more characterisation of F worth recording, because it is the statistical-mechanics form of the variational principle that underlies both the ELBO of diffusion models (Chapter 4) and the Bethe/BP free energies (Chapter 7). For *any* trial density μ,

$$\underbrace{\mathbb{E}_\mu[H] - \tfrac{1}{\beta} S[\mu]/k_B}_{\text{variational free energy } F[\mu]} \;=\; F \;+\; \tfrac{1}{\beta}\, D_{\mathrm{KL}}(\mu \,\|\, \rho_\beta) \;\geq\; F, \tag{2.11}$$

with equality iff $\mu = \rho_\beta$: the Boltzmann distribution is the unique minimiser of the variational free energy, and the gap is exactly a Kullback–Leibler divergence. Every variational method in machine learning is a restriction of (2.11) to a tractable family of μ's.

2.5 Interacting systems: the Ising paradigm

Everything so far holds for arbitrary H; the physics (and the difficulty) enters when the energy couples many degrees of freedom. The canonical example is the *Ising model* (Ising, 1925): binary spins $\sigma_i \in \{-1, +1\}$ on the nodes of a graph $G = (V, \mathcal{E})$, with energy

$$H(\sigma) \;=\; -\sum_{(i,j)\in\mathcal{E}} J_{ij}\, \sigma_i \sigma_j \;-\; \sum_{i\in V} h_i\, \sigma_i, \tag{2.12}$$

where J_{ij} are pairwise couplings and h_i external fields. The Gibbs distribution $p(\sigma) \propto e^{-\beta H(\sigma)}$ makes neighbouring spins prefer alignment when $J_{ij} > 0$. Three lessons from this model shape the rest of the thesis.

(i) The partition function is the hard part. $Z = \sum_\sigma e^{-\beta H(\sigma)}$ runs over $2^{|V|}$ configurations. On two-dimensional lattices Onsager computed it exactly (Onsager, 1944); on general graphs the computation is #P-hard, and *all* of approximate inference — MCMC (Chapter 3), variational methods, message passing (Chapter 7) — exists because of this wall. On *trees*, however, Z and all marginals are computable exactly in linear time by the transfer-matrix/belief-propagation recursion; this tractable island is precisely where our research model lives.

(ii) Collective order emerges. For β beyond a critical value the two-dimensional model magnetises spontaneously: a macroscopic fraction of spins align even at $h = 0$, and the symmetric state splits into two ergodic components. This spontaneous symmetry breaking is the archetype of a *phase transition*, and its dynamical counterpart — a generation trajectory committing to one mode of a multimodal distribution — is exactly the *speciation* phenomenon of diffusion models (Biroli et al., 2024; Raya and Ambrogioni, 2023).

(iii) Disorder creates complexity. Making the couplings J_{ij} random (of both signs) yields spin glasses — and, as the next sections recount, spin glasses yielded neural networks. This is where the physics of this chapter stops being background and becomes the literature this thesis belongs to.

2.6 Disorder, frustration, and spin glasses

In a ferromagnet all couplings agree, and the two ground states (all spins up, all spins down) are found by inspection. Draw the couplings J_{ij} at random with both signs (Edwards and Anderson, 1975; Sherrington and Kirkpatrick, 1975) and this transparency collapses. A triangle with two positive bonds and one negative bond already cannot satisfy all three: whatever the spins do, at least one bond pays. This is *frustration*, and its large-N consequence is a free-energy landscape shattered into exponentially many valleys separated by extensive barriers — the system has many near-equally-good ways of being, none of them readable off the couplings. The mean-field theory of this phenomenon, developed through the replica and cavity methods (Mézard et al., 1987), and the Thouless–Anderson–Palmer (TAP) self-consistency equations (Thouless et al., 1977), took a decade to control and earned Parisi the 2021 Nobel Prize in Physics.

Two exports from this episode matter here. First, a language: rugged landscapes, metastable states, order parameters that measure similarity rather than magnetisation. Machine learning speaks this language every time it discusses loss surfaces and local minima. Second, a technology: the cavity and TAP equations are recursions on *local* quantities that solve a *global* inference problem, and they are the direct ancestors of the message-passing algorithms — belief propagation and its approximations (Mézard and Montanari, 2009; Zdeborová and Krzakala, 2016) — that Chapter 7 deploys on diffusion scores. The path from disordered magnets to algorithms is not an analogy; it is a genealogy.

2.7 The Hopfield network: memory as a phase of matter

Hopfield (1982) asked a physicist's question about memory: can a system of simple binary units *store* and *retrieve* patterns as a collective effect, the way a magnet stores a magnetisation? His construction is an Ising system (2.12) in disguise. Take N binary neurons $\sigma_i \in \{-1, +1\}$ and P patterns $\xi^\mu \in \{-1, +1\}^N$ to be memorised, and wire the couplings by the *Hebb rule* — neurons that fire together wire together (Hebb, 1949):

$$J_{ij} \;=\; \frac{1}{N}\sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu, \qquad H(\sigma) \;=\; -\frac{1}{2}\sum_{i\neq j} J_{ij}\,\sigma_i\sigma_j. \tag{2.13}$$

The dynamics is the simplest imaginable: pick a neuron, align it with its local field, repeat,

$$\sigma_i \;\leftarrow\; \operatorname{sign}\Big(\textstyle\sum_{j\neq i} J_{ij}\,\sigma_j\Big). \tag{2.14}$$

Toolbox 2.5 — The Hopfield dynamics descends the energy

Let $h_i = \sum_{j\neq i} J_{ij}\sigma_j$ be the local field on neuron i and suppose the update (2.14) flips it, $\sigma_i' = -\sigma_i$. Only the terms of H containing σ_i change, so

$$\Delta H \;=\; H(\sigma') - H(\sigma) \;=\; -\,(\sigma_i' - \sigma_i)\,h_i \;=\; -\,2\,\sigma_i'\,h_i.$$

The update sets $\sigma_i' = \operatorname{sign}(h_i)$, hence $\sigma_i' h_i = |h_i| \geq 0$ and $\Delta H \leq 0$, with strict decrease whenever a flip actually occurs. Since H is bounded below on a finite configuration space, the dynamics must terminate in a local minimum: *every* initial condition falls into an attractor. Memory retrieval is this fall: initialise at a corrupted pattern, descend, and read off the fixed point. □

The patterns are, by construction, near the bottoms of the energy (2.13): retrieval works because each memory is an attractor with a basin. How many memories fit is a genuine phase-transition question, answered with spin-glass tools by Amit et al. (1985): for random patterns with $P = \alpha N$, retrieval survives only below a critical load $\alpha_c \approx 0.138$, beyond which the memories are crushed into a useless glassy phase. Memory, in this model, is literally a phase of matter, entered and left through sharp transitions. The single-pattern case $P = 1$ — known as the Mattis model (Mattis, 1976) — is gauge-equivalent to the uniform ferromagnet: flipping $\sigma_i \to \xi_i \sigma_i$ maps the stored pattern onto the all-up state. This innocent observation should be kept in mind: the Mattis model returns in Section 4.8 as the exactly solvable "teacher" in a modern analysis of how energy-based models *learn*.

In 2024 the Nobel Prize in Physics went to John Hopfield and Geoffrey Hinton "for foundational discoveries and inventions that enable machine learning with artificial neural networks" (The Royal Swedish Academy of Sciences, 2024) — formal recognition that the lineage traced in this section is physics, not just inspired by it.

2.8 From Hopfield to Boltzmann machines: learning enters

The Hopfield network stores what its designer wires into it. The decisive step towards machine *learning* was to make the couplings themselves the unknowns. Promote the zero-temperature dynamics (2.14) to stochastic updates at temperature $T = 1/\beta$ (each neuron aligns with its field only with probability given by the Boltzmann factor); the stationary law of this dynamics is exactly the Gibbs measure $p(\sigma) \propto e^{-\beta H(\sigma)}$ of the Ising energy. A *Boltzmann machine* (Ackley et al., 1985) is this object run backwards: fix a dataset, and ask for the couplings J that make the Gibbs measure reproduce it. Gradient ascent on the log-likelihood gives a rule of striking symmetry,

$$\frac{\partial}{\partial J_{ij}} \mathbb{E}_{\text{data}}\Big[\log p(\sigma)\Big] \;=\; \beta\left(\underbrace{\langle\sigma_i\sigma_j\rangle_{\text{data}}}_{\text{positive phase}} - \underbrace{\langle\sigma_i\sigma_j\rangle_{\text{model}}}_{\text{negative phase}}\right), \tag{2.15}$$

learning stops when the model's correlations match the data's. This is the maximum-entropy principle of Toolbox 2.3 turned into an algorithm: matching prescribed moments with an exponential-family measure. The rule also contains, in miniature, the central obstruction of the field. The positive phase is an average over data — cheap. The negative phase is an average over the *model's* Gibbs measure — a partition-function computation, the wall of Section 2.5, approachable only through the sampling methods of Chapter 3.

The *restricted* Boltzmann machine (RBM) tames this cost by architecture (Smolensky, 1986): split the units into a visible layer v and a hidden layer h with couplings only across layers, $E(v,h) = -v^{\mathsf{T}}Wh - b^{\mathsf{T}}v - c^{\mathsf{T}}h$. On this bipartite graph each layer is conditionally independent given the other, so Gibbs sampling alternates two exact, fully parallel steps, and the contrastive-divergence approximation to (2.15) made training practical (Hinton, 2002). Stacked RBMs — deep belief networks — delivered the first working deep architectures and relaunched neural networks as a field (Hinton et al., 2006). The lineage then split. The supervised branch, powered by backpropagation (Rumelhart et al., 1986), scaled through convolutional networks to the modern deep-learning era (Krizhevsky et al., 2012; LeCun et al., 2015) and largely left equilibrium physics behind. The *generative* branch — the one this thesis lives in — kept the Gibbsian core: a model is an energy, learning is moment matching, sampling is dynamics. Chapter 4 follows that branch from energy-based models to diffusion.

2.9 The statistical mechanics of learning

Physics did not stop contributing machines; it also built the theory of their behaviour. Gardner (1988) showed that learning itself can be treated as an equilibrium problem:

fix a set of examples and compute, by replica methods, the volume of coupling space compatible with them — capacity, generalisation, and their transitions then come out as thermodynamics in *weight* space. This programme, extended through teacher–student settings, matured into the modern statistical physics of inference (Zdeborová and Krzakala, 2016), whose central lesson recurs throughout this thesis: high-dimensional inference problems have phases, and algorithms fail or succeed at sharp boundaries.

Two strands of this literature frame our research questions directly. The first is algorithmic: Mézard (2017) derived the mean-field message-passing (TAP) equations of the Hopfield model and its generalisations, exhibiting the equivalence between Hopfield networks and RBMs and showing that retrieval can be organised as a message-passing computation — the same machinery, transplanted from memories to diffusion scores, that Chapter 7 runs on our model. The second is dynamical: training itself undergoes phase transitions. In linear networks, gradient descent learns the singular modes of the data covariance sequentially (Saxe et al., 2014); in RBMs, the weight matrix's singular vectors align with the data's principal components in the early phase of learning (Decelle et al., 2017), and hidden units develop compositional receptive fields through condensation transitions (Tubiana and Monasson, 2017). The sharpest result in this line — Bachtis et al. (2024), examined in depth in Section 4.8 — shows that RBM training is an entire *cascade* of second-order phase transitions, one per learned feature. Learning, like memory, turns out to be a sequence of symmetry breakings.

2.10 What this thesis takes from physics

A *complex system* is, operationally, a system whose macroscopic behaviour is not readable off its microscopic rules: interactions generate collective phenomena — order, phases, transitions, slow dynamics — that must be derived, not postulated. Statistical mechanics is the toolbox for that derivation, and after this chapter's story its objects map one-to-one onto the objects of modern generative modelling:

Statistical mechanics	**Generative modelling**
microstate x	data point / latent configuration
energy $H(x)$	negative log-density $-\log p(x)$ (up to Z)
Boltzmann distribution $e^{-\beta H}/Z$	model distribution
partition function Z	normalising constant / evidence
free energy F	negative ELBO (up to constants)
Hebbian couplings, attractors	stored memories, retrieval
moment matching (2.15)	maximum-likelihood training
Langevin dynamics	sampling / reverse diffusion
phase transition	feature condensation, speciation
cavity / TAP equations	belief propagation / AMP

The next chapter supplies the probabilistic dynamics — Markov chains and their continuous-time limits — that turn the static ensembles of this chapter into processes that *move* probability around, which is what a diffusion model ultimately is.

Chapter 3

Stochastic Processes, SDEs, and the Ornstein–Uhlenbeck Channel

Chapter 2 answered the question *"which distribution?"* — Boltzmann's. This chapter answers *"how does probability move?"*, in discrete and in continuous time, and it earns its place in the thesis three times over. Markov chains are the *data* whose diffusion score we compute exactly (the AR(1) sequence of Section 3.2 is the running example of every research chapter); stochastic differential equations are the *corruption and generation processes* of diffusion models (the Ornstein–Uhlenbeck channel of Section 3.5 appears on every page of Chapters 5–7); and the Fokker–Planck and time-reversal theorems of Sections 3.6–3.7 are the mathematical licence for the entire generative programme reviewed in Chapter 4. Everything proved here is used later; nothing is included for completeness' sake. Standard references are Robert and Casella (2004); Brockwell and Davis (1991); Øksendal (2003); Gardiner (2009); Risken (1996).

3.1 Markov chains: kernels, stationarity, reversibility

Definition 3.1 (Markov chain)**.** A sequence of random variables $(X_k)_{k\geq 0}$ with values in a state space $\mathcal{X}$ is a *Markov chain* if for every k and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}\Big(X_{k+1} \in A \,\Big|\, X_k, X_{k-1}, \dots, X_0\Big) \;=\; \mathbb{P}\Big(X_{k+1} \in A \,\Big|\, X_k\Big). \tag{3.1}$$

The conditional law is encoded by a *transition kernel* $M(x' \mid x)$: a density (or matrix, for discrete $\mathcal{X}$) with $\int M(x'|x)\,\mathrm{d}x' = 1$ for every x.

The Markov property (3.1) is exactly a statement about *factorisation*: the joint law of the first K states is

$$p(x_0, x_1, \dots, x_{K-1}) \;=\; p_0(x_0) \prod_{k=0}^{K-2} M(x_{k+1} \mid x_k). \tag{3.2}$$

This one-line factorisation is the single most important equation of the thesis. It says that the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables — the structure that makes the chain a *tree* in the factor-graph sense of Chapter 7, that makes its precision matrix tridiagonal in the Gaussian case of Chapter 5, and that makes belief propagation exact.

Marginal distributions evolve by integrating the kernel: $p_{k+1}(x') = \int M(x'|x)\, p_k(x)\,\mathrm{d}x$, compactly $p_{k+1} = Mp_k$. A distribution π is *stationary* (invariant) for M if $\pi = M\pi$; a chain started in π stays in π forever, and the trajectory is then a *stationary process*. The kernel satisfies *detailed balance* with respect to π if

$$\pi(x)\, M(x' \mid x) \;=\; \pi(x')\, M(x \mid x') \qquad \text{for all } x, x' : \tag{3.3}$$

the equilibrium probability flux from x to x' equals the reverse flux, so the chain, watched in equilibrium, is statistically indistinguishable run forwards or backwards. Detailed balance implies stationarity (integrate (3.3) over x), and its power is practical: it is a *local,* *pointwise* condition, checkable without computing any integral.

Stationarity alone does not guarantee that the chain *finds* π; for that one needs the chain to be able to reach everywhere (*irreducibility*) without getting trapped in cycles (*aperiodicity*). Under these conditions the fundamental convergence theorem holds (Robert and Casella, 2004): the marginal law of X_k converges to π from any initial condition, and time averages converge to expectations under π (the ergodic theorem),

$$\frac{1}{N}\sum_{k=1}^{N} f(X_k) \xrightarrow[N\to\infty]{\text{a.s.}} \mathbb{E}_\pi[f]. \tag{3.4}$$

Equation (3.4) is the rigorous form of the ergodic postulate invoked in Section 2.3, with the physical dynamics replaced by an algorithmic one.

3.2 The AR(1) chain: the running example of the thesis

We now introduce the specific Markov chain that the research chapters solve completely: the first-order autoregressive (AR(1)) process, the canonical Gaussian Markov chain (Brockwell and Davis, 1991). Let $\alpha \in (-1, 1)$ and let (η_k) be i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$ innovations; define

$$a_{k+1} \;=\; \alpha\, a_k + \eta_k, \qquad k = 0, 1, \ldots, K-2. \tag{3.5}$$

The transition kernel is Gaussian, $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, and the joint law of the trajectory follows from (3.2). Everything about this chain is computable in closed form; the next toolbox derives the three facts used throughout.

Toolbox 3.1 — Stationary distribution and autocovariance of AR(1)

(i) Stationary variance. Suppose $a_k \sim \mathcal{N}(0, \sigma^2)$ and impose that a_{k+1} has the same variance. Since η_k is independent of a_k,

$$\mathrm{Var}(a_{k+1}) = \alpha^2 \,\mathrm{Var}(a_k) + \sigma_\eta^2 \overset{!}{=} \mathrm{Var}(a_k) \quad \Longrightarrow \quad \sigma_\infty^2 = \frac{\sigma_\eta^2}{1-\alpha^2}.$$

The fixed point exists precisely because $|\alpha| < 1$; the map $\sigma^2 \mapsto \alpha^2\sigma^2 + \sigma_\eta^2$ is a contraction, so *any* initial variance converges to σ_∞^2 geometrically. Since Gaussianity is preserved by the affine recursion, the stationary law is $\pi = \mathcal{N}(0, \sigma_\infty^2)$.

(ii) Autocovariance. In the stationary regime, for $d \geq 1$, unroll the recursion d times: $a_{k+d} = \alpha^d a_k + \sum_{j=0}^{d-1} \alpha^j \eta_{k+d-1-j}$. All innovation terms are independent of a_k, hence

$$\mathrm{Cov}(a_k, a_{k+d}) = \alpha^d \,\mathrm{Var}(a_k) = \alpha^d \,\sigma_\infty^2,$$

and by symmetry $\mathrm{Cov}(a_i, a_j) = \alpha^{|i-j|}\sigma_\infty^2$ for all i, j: correlations decay *geometrically* with the lag, with rate α. The covariance matrix of K consecutive frames,

$$(\Sigma_0)_{ij} = \sigma_\infty^2 \,\alpha^{|i-j|},$$

depends on (i, j) only through $i - j$: it is a symmetric *Toeplitz* matrix, a structure we exploit heavily in Chapter 5.

(iii) Detailed balance. With $\pi = \mathcal{N}(0, \sigma_\infty^2)$ and $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, both sides of (3.3) are proportional to

$$\exp\left[-\frac{a^2}{2\sigma_\infty^2} - \frac{(a' - \alpha a)^2}{2\sigma_\eta^2}\right] = \exp\left[-\frac{a^2 + (a')^2 - 2\alpha\, a a'}{2\sigma_\eta^2}\right],$$

where the exponent on the right is *symmetric* in (a, a'). Hence the stationary AR(1) chain is reversible: statistically, the movie of the chain looks the same played in either direction. □

The AR(1) chain is thus the perfect miniature of everything this chapter discusses: a

Markov kernel with an explicit stationary law, geometric convergence, reversibility, and — decisive for us — *exact Gaussian algebra* at every step. When in Chapter 5 we corrupt each frame of this chain with independent noise and ask for the joint score of the result, every quantity will remain in closed form.

3.3 Sampling by dynamics: MCMC and Langevin

Sampling inverts the logic of the previous section. There, a kernel M was given and we asked for its stationary law π. In sampling problems we are given π — typically a Boltzmann distribution $\pi(x) \propto e^{-\beta H(x)}$ known only up to its partition function — and we must *design* a kernel whose stationary law is π; the ergodic theorem (3.4) then converts one long trajectory into expectation values. This is how the negative phase of Boltzmann-machine learning (2.15) is estimated in practice, and the idea originates, fittingly, in computational statistical mechanics (Metropolis et al., 1953).

The *Metropolis–Hastings* chain (Metropolis et al., 1953; Hastings, 1970) moves from x by proposing $x' \sim g(\cdot|x)$ from any tractable proposal kernel and accepting the move with probability

$$A(x \to x') \;=\; \min\left\{1,\; \frac{\pi(x')\, g(x\,|\,x')}{\pi(x)\, g(x'\,|\,x)}\right\}, \tag{3.6}$$

staying at x otherwise. The target enters only through the *ratio* $\pi(x')/\pi(x)$, so the intractable partition function cancels: MH samples from $e^{-\beta H}/Z$ while only ever evaluating energy differences. The acceptance rule is engineered so that the resulting kernel satisfies detailed balance (3.3) with respect to π; the short verification is Derivation B.1 in Appendix B. When x is high-dimensional but each conditional $\pi(x_i|x_{\setminus i})$ is tractable, the *Gibbs sampler* (Geman and Geman, 1984) cycles through the coordinates resampling each from its conditional — an MH move whose acceptance probability is identically one. Gibbs sampling thrives exactly when the model has sparse conditional structure: for a Markov chain, $\pi(x_k|x_{\setminus k})$ depends only on the two neighbours, and for the bipartite RBM of Section 2.8 each layer updates in one parallel step. The reader should hear, already, the theme of Chapter 7: *inference cost follows graph structure.*

For continuous x and differentiable energy, a different design principle uses gradients. Consider the discrete-time update

$$x_{k+1} \;=\; x_k + \epsilon \, \nabla \log \pi(x_k) \;+\; \sqrt{2\epsilon}\, \xi_k, \qquad \xi_k \sim \mathcal{N}(0, I), \tag{3.7}$$

the (unadjusted) *Langevin algorithm*. The drift pushes towards high probability; the noise maintains exploration. Two observations preview what follows. First, (3.7) requires only the *score* $\nabla \log \pi$ — the normalising constant is never needed; this is precisely why the score is the natural object for generative modelling (Chapter 4), and why so much of this thesis is devoted to computing scores exactly. Second, as $\epsilon \to 0$ the recursion (3.7) becomes a stochastic differential equation — the Langevin diffusion — whose stationary law is exactly π. Making that limit precise requires the Itô calculus and Fokker–Planck theory that occupy the rest of the chapter, where the claim is verified by direct computation.

3.4 From ODEs to SDEs: the Itô calculus

An ordinary differential equation $\dot{x} = f(x, t)$ moves a point along a deterministic velocity field. Physical systems, however, feel rapid, unresolved forces — collisions, thermal agitation — whose cumulative effect is well modelled by adding noise:

$$\mathrm{d}X_t \;=\; \underbrace{f(X_t, t)\,\mathrm{d}t}_{\text{drift}} \;+\; \underbrace{g(t)\,\mathrm{d}W_t}_{\text{diffusion}}. \tag{3.8}$$

Here W_t is a *Wiener process* (standard Brownian motion): the unique process with $W_0 = 0$, independent increments, $W_t - W_s \sim \mathcal{N}(0, t - s)$, and continuous paths. Equation (3.8) is shorthand for the integral equation $X_t = X_0 + \int_0^t f(X_s, s)\,\mathrm{d}s + \int_0^t g(s)\,\mathrm{d}W_s$, and the whole subtlety lives in the last term: Brownian paths are nowhere differentiable, so $\int g\,\mathrm{d}W$ cannot be a Riemann–Stieltjes integral. Itô's construction (Itô, 1944) defines it as the limit of left-endpoint sums,

$$\int_0^t g(s)\,\mathrm{d}W_s \;=\; \lim_{n\to\infty} \sum_{j=0}^{n-1} g(s_j)\,\Big(W_{s_{j+1}} - W_{s_j}\Big), \tag{3.9}$$

where the evaluation at the *left* endpoint s_j makes the integrand independent of the coming increment. Two consequences follow immediately: the Itô integral is a martingale (zero mean), and its variance is given by the *Itô isometry*, $\mathbb{E}\Big[(\int_0^t g\,\mathrm{d}W)^2\Big] = \int_0^t g(s)^2\,\mathrm{d}s$, both because cross terms $\mathbb{E}[g(s_j)(W_{s_{j+1}} - W_{s_j})]$ vanish by independence.

The price of the construction is a new chain rule. Its origin is the *quadratic variation* of Brownian motion: increments scale as $\mathrm{d}W_t \sim \sqrt{\mathrm{d}t}$, so second-order terms in $\mathrm{d}W$ are first-order in $\mathrm{d}t$ and cannot be discarded.

Toolbox 3.2 — Itô's lemma, and where the extra term comes from

Let $Y_t = \phi(X_t, t)$ with ϕ twice differentiable and X_t solving (3.8). Taylor-expand ϕ to second order:

$$\mathrm{d}Y = \frac{\partial\phi}{\partial t}\,\mathrm{d}t + \frac{\partial\phi}{\partial x}\,\mathrm{d}X + \frac{1}{2}\frac{\partial^2\phi}{\partial x^2}\,(\mathrm{d}X)^2 + \dots$$

Substitute $\mathrm{d}X = f\,\mathrm{d}t + g\,\mathrm{d}W$ and expand the square:

$$(\mathrm{d}X)^2 = f^2(\mathrm{d}t)^2 + 2fg\,\mathrm{d}t\,\mathrm{d}W + g^2(\mathrm{d}W)^2.$$

Now use the multiplication table of Itô calculus, which is nothing but the scaling $\mathrm{d}W \sim \sqrt{\mathrm{d}t}$ made systematic:

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\,\mathrm{d}W = 0, \qquad (\mathrm{d}W)^2 = \mathrm{d}t.$$

The last identity is the rigorous statement that the quadratic variation of W over $[0, t]$ equals t: for a partition of mesh $\to 0$, $\sum_j (W_{s_{j+1}} - W_{s_j})^2 \to t$ in mean square, since the sum has mean $\sum_j (s_{j+1} - s_j) = t$ and variance $2\sum_j (s_{j+1} - s_j)^2 \to 0$. Keeping the surviving terms:

$$\mathrm{d}Y = \left(\frac{\partial\phi}{\partial t} + f\,\frac{\partial\phi}{\partial x} + \frac{g^2}{2}\,\frac{\partial^2\phi}{\partial x^2}\right)\mathrm{d}t \;+\; g\,\frac{\partial\phi}{\partial x}\,\mathrm{d}W.$$

The half-Laplacian term — absent from the classical chain rule — is the engine of everything that follows: it is why densities *spread*, why the Fokker–Planck equation has a second-order term, and why exp-transformations of SDEs pick up variance corrections.

□

3.5 The Ornstein–Uhlenbeck process: the corruption channel

The noising process used by diffusion models — and by every result of this thesis — is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled back to the origin by a linear spring,

$$\mathrm{d}X_t \;=\; -X_t\,\mathrm{d}t \;+\; \sqrt{2}\,\mathrm{d}W_t, \qquad X_0 = a. \tag{3.10}$$

(The drift and diffusion constants are a normalisation choice — this is the "variance-preserving" convention of the diffusion literature (Song et al., 2021); any other choice rescales time.) The OU process is the unique Gaussian, Markov, stationary process in continuous time, and it is exactly solvable.

Toolbox 3.3 — Exact solution of the OU SDE by integrating factor

Multiply (3.10) by e^t and consider $Y_t = e^t X_t$. By Itô's lemma with $\phi(x,t) = e^t x$ (here $\partial^2\phi/\partial x^2 = 0$, so no correction term arises):

$$\mathrm{d}Y_t = e^t X_t\,\mathrm{d}t + e^t\Big(-X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t\Big) = \sqrt{2}\,e^t\,\mathrm{d}W_t.$$

The drift cancels exactly — that is what the integrating factor is for. Integrate from 0 to t and undo the substitution:

$$X_t \;=\; a\,e^{-t} \;+\; \sqrt{2}\int_0^t e^{-(t-s)}\,\mathrm{d}W_s.$$

The integral is Gaussian (limit of Gaussian sums), with mean zero and variance given by the Itô isometry:

$$\mathrm{Var}(X_t) = 2\int_0^t e^{-2(t-s)}\,\mathrm{d}s = 2\,e^{-2t}\,\frac{e^{2t}-1}{2} = 1-e^{-2t}.$$

Hence the *transition kernel* of the OU channel is

$$p(x_t \,|\, X_0 = a) \;=\; \mathcal{N}\Big(x_t;\; a\,e^{-t},\; \Delta_t\Big), \qquad \boxed{\Delta_t \;=\; 1-e^{-2t}}.$$

□

The boxed kernel is used on every page of Chapters 5 –7, so we fix its reading now. Corrupting a data point a for a time t produces

$$x \;=\; a\,e^{-t} \;+\; \sqrt{\Delta_t}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1): \tag{3.11}$$

the signal is *contracted* by e^{-t} while isotropic noise of variance Δ_t fills in. At $t = 0$: $x = a$ (no corruption); as $t \to \infty$: $x \sim \mathcal{N}(0,1)$ regardless of a (all information destroyed, the equilibrium of the spring). The signal-to-noise ratio e^{-2t}/Δ_t sweeps continuously from ∞ to 0: the OU channel interpolates between the data distribution and a standard Gaussian, which is precisely what a diffusion model needs.

Two structural remarks, both load-bearing later. First, the OU kernel is Gaussian *in a as well as in x*: viewed as a function of the clean variable, $\mathcal{N}(x; ae^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - xe^t)^2]$ — a Gaussian "likelihood" factor. This is the fact that makes the posteriors of Chapter 5 jointly Gaussian and the messages of Chapter 7 closable. Second, if a random vector $a \sim \mathcal{N}(\mu, \Sigma_0)$ is pushed through (3.11) coordinate-wise with *independent* noises, the output is Gaussian with

$$\mu_t = e^{-t}\mu, \qquad \Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I, \tag{3.12}$$

since means and covariances transform linearly and the added noise is white. Equation (3.12) — shrink the covariance, add a multiple of the identity — is the entire "forward process" of this thesis in one line.

3.6 The Fokker–Planck equation: transporting densities

The SDE (3.8) moves *samples*; the corresponding motion of the *density* $p_t(x)$ is governed by the Fokker–Planck (or forward Kolmogorov) equation. This change of viewpoint — from trajectories to densities — is the same one Liouville's theorem performed for Hamiltonian flow, and the derivation is parallel, with the Itô correction supplying a new, diffusive term.

Toolbox 3.4 — Derivation of the Fokker–Planck equation

Let ϕ be an arbitrary smooth test function of compact support, and compute $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\phi(X_t)]$ in two ways.

Way 1: through the samples. By Itô's lemma,

$$\mathrm{d}\phi(X_t) = \Big(f\,\phi' + \tfrac{g^2}{2}\,\phi''\Big)\mathrm{d}t + g\,\phi'\,\mathrm{d}W_t,$$

and taking expectations kills the martingale term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}[\phi(X_t)] = \mathbb{E}\Big[f(X_t,t)\,\phi'(X_t)\Big] + \frac{g(t)^2}{2}\,\mathbb{E}\Big[\phi''(X_t)\Big].$$

Way 2: through the density. $\mathbb{E}[\phi(X_t)] = \int \phi(x)\, p_t(x)\, \mathrm{d}x$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\, \mathbb{E}[\phi(X_t)] = \int \phi(x)\, \frac{\partial p_t(x)}{\partial t}\, \mathrm{d}x.$$

Equate the two, and integrate Way 1 by parts to move all derivatives off ϕ (boundary terms vanish by compact support): once for the drift term, twice for the diffusion term,

$$\int \phi\, \frac{\partial p_t}{\partial t} = \int \phi \left[-\frac{\partial}{\partial x}\big(f\, p_t\big) + \frac{g^2}{2}\, \frac{\partial^2 p_t}{\partial x^2} \right] \mathrm{d}x.$$

Since ϕ is arbitrary, the integrands must agree pointwise:

$$\boxed{\frac{\partial p_t}{\partial t} = -\frac{\partial}{\partial x}\Big(f(x,t)\, p_t(x)\Big) + \frac{g(t)^2}{2}\, \frac{\partial^2 p_t(x)}{\partial x^2}.}$$

The first term transports probability along the drift (as in Liouville); the second spreads it diffusively — it is the macroscopic footprint of the Itô correction. □

3.6.1 Stationary law of Langevin dynamics: closing the loop

Apply the Fokker–Planck equation to the *Langevin SDE* associated with a potential U (the continuous-time limit of the algorithm (3.7)):

$$\mathrm{d}X_t = -\nabla U(X_t)\, \mathrm{d}t + \sqrt{2\beta^{-1}}\, \mathrm{d}W_t. \tag{3.13}$$

A stationary density solves $0 = \nabla \cdot (\nabla U\, p + \beta^{-1} \nabla p)$, i.e. the probability current $J = -(\nabla U)\, p - \beta^{-1} \nabla p$ is divergence-free; imposing the natural (decaying) boundary conditions forces $J \equiv 0$, hence $\nabla p / p = -\beta \nabla U$, hence

$$p_\infty(x) \;\propto\; e^{-\beta U(x)}\;: \tag{3.14}$$

Langevin dynamics equilibrates to the Boltzmann distribution of Chapter 2. This is the two-line proof of the claim left pending in Section 3.3, and it explains the deep kinship of sampling and physics: a Langevin sampler *is* an overdamped physical system at temperature $1/\beta$. Note also that the drift of (3.13) is (up to β) the *score* of the target, $-\nabla U = \nabla \log p_\infty$: dynamics that sample a distribution are driven by its score.

For the OU process, $U(x) = x^2/2$ and $\beta = 1$: the stationary law is $\mathcal{N}(0,1)$, confirming the $t \to \infty$ limit of Section 3.5.

3.7 Time reversal: the mother theorem of diffusion models

One last piece of SDE theory converts everything above into a generative method. Suppose data $X_0 \sim p_0$ are corrupted by a forward SDE (3.8) up to time T, producing marginals p_t. Can the corruption be undone — can we find an SDE which, run from $X_T \sim p_T$ backwards, reproduces the same marginals? The answer (Anderson, 1982) is yes, and the reverse drift needs exactly one unknown object:

$$\mathrm{d}X_t \;=\; \Big[f(X_t,t) \;-\; g(t)^2\,\nabla_x \log p_t(X_t)\Big]\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}\bar{W}_t, \tag{3.15}$$

where time runs backwards from T to 0 and $\bar{W}$ is a reverse-time Wiener process. The correction to the drift is the *score* $\nabla_x \log p_t$ of the noised marginal — the object defined in (1.1) and computed exactly, for Markov-chain data, in Chapters 5–7. A heuristic that makes (3.15) plausible follows from the Fokker–Planck equation: rewriting its diffusion term as a transport term,

$$\frac{g^2}{2}\frac{\partial^2 p_t}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{g^2}{2}\,p_t\,\frac{\partial \log p_t}{\partial x}\right),$$

one sees that the density evolution of (3.8) is *identical* to that of the noise-free ODE with velocity $f - \frac{g^2}{2}\nabla \log p_t$ (the *probability-flow ODE* of Song et al., 2021); running that transport backwards, and re-injecting noise consistently, yields (3.15). The rigorous statement and proof are in Anderson (1982).

3.8 From algorithmic chains to data chains

A closing change of perspective, to set up the research problem. In this chapter Markov chains began as *algorithms*: their stationary law was the target, their trajectory a means to an end. From Chapter 5 onwards the chain (3.5) plays the opposite role: it is the *data-generating process*, its trajectory $(a_0, \ldots, a_{K-1})$ is one data point — a "video" of K frames — and the object of interest is the probability law of the *whole trajectory* after each frame has been independently corrupted by the OU channel (3.11). The factorisation (3.2) then stops being a computational convenience and becomes the *structure to be discovered*: the entire research programme of this thesis can be phrased as asking how much of (3.2) survives noising, and how an inference algorithm — or a neural network — can exploit what survives. In one sentence, the chapter's summary: the OU channel destroys data at a known rate, the Fokker–Planck equation describes the destruction at the level of densities, and Anderson's theorem says the destruction is invertible by anyone who knows the score $\nabla \log p_t$ — which is why the rest of this thesis is about computing that score, exactly, when the data have Markov structure.

Chapter 4

From Energy-Based Models to Diffusion and Flows

Chapter 2 left the generative branch of machine learning at the Boltzmann machine: a model is an energy, learning is moment matching, sampling is dynamics. This chapter follows that branch to the present. It reviews, in the depth their role in this thesis demands, the three ideas the research chapters build on: the energy-based view of modelling (LeCun et al., 2006), the score-matching principle that removed the partition function from training (Hyvärinen, 2005; Vincent, 2011), and the diffusion construction that turned denoising into a complete generative method (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021) — together with its deterministic sibling, the flow model. The chapter then reviews the statistical-physics literature on diffusion models (Biroli and Mézard, 2023; Biroli et al., 2024; Ghio et al., 2024) and, on the training side, the cascade of phase transitions in energy-based-model learning (Bachtis et al., 2024), and closes by stating the gap this thesis fills.

4.1 Energy-based models

Modern generative modelling asks a single question: given samples of an unknown distribution p_0, build a machine that produces new ones. The last decade produced several answers — variational autoencoders (Kingma and Welling, 2014), adversarial networks (Goodfellow et al., 2014), normalizing flows (Rezende and Mohamed, 2015), diffusion models — and it is easy to present them as unrelated tricks. The energy-based view of LeCun et al. (2006) is the antidote: it is the direct continuation of the Boltzmann-machine programme of Section 2.8, and, as this chapter unfolds, diffusion models will appear not as a new species but as the member of the family that finally solved its oldest problem.

An *energy-based model* (EBM) over $x \in \mathcal{X}$ assigns to each configuration a scalar energy $E_\theta(x)$ measuring how badly x fits the model's idea of the data — low energy for plausible configurations, high for implausible ones. In LeCun et al.'s formulation the energy is the primitive object: inference means finding low-energy configurations, and learning means *shaping the landscape* — pushing the energy of observed data down and the energy of everything else up. A probabilistic model is recovered whenever one normalises with the Gibbs measure of Chapter 2,

$$p_\theta(x) \;=\; \frac{e^{-E_\theta(x)}}{Z(\theta)}, \qquad Z(\theta) = \int_{\mathcal{X}} e^{-E_\theta(x)}\,\mathrm{d}x, \tag{4.1}$$

the Boltzmann distribution (2.9) with β absorbed into the energy and the physics replaced by a parametric family. The appeal of (4.1) is that E_θ is unconstrained — any function defines a valid unnormalised density; the price is the partition function. Every basic operation hits it. *Evaluating* $p_\theta(x)$ requires $Z(\theta)$, intractable in general (Section 2.5). *Sampling* requires the MCMC machinery of Section 3.3, whose mixing time can be exponential in the presence of metastability. *Learning* by maximum likelihood requires the gradient

$$-\nabla_\theta \log p_\theta(x) = \nabla_\theta E_\theta(x) - \mathbb{E}_{p_\theta}\!\left[\nabla_\theta E_\theta\right], \tag{4.2}$$

the "push down on data, pull up on the model's own samples" rule — the continuous-variable form of the Boltzmann-machine rule (2.15) — whose second term again demands sampling from p_θ. LeCun et al. (2006) frame the alternatives systematically: either fight the negative phase with *contrastive* losses that pull up only on well-chosen offending configurations (contrastive divergence (Hinton, 2002) being the MCMC-flavoured member of the family), or design architectures and regularisers that bound the volume of low-energy space so that no pull-up is needed. The lesson of the tutorial is that the partition function is not an accounting detail; it is *the* design constraint of the field.

There are two classical escape routes, and this thesis walks both. The first is the *score*: since $\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$, the gradient of the log-density in x is free of Z — differentiating in x kills the constant. Sections 4.4–4.6 build the modern edifice standing on this observation. The second is *structure*: when the energy decomposes into local terms, marginals and even Z itself become computable by dynamic programming; that route — Markov random fields, factor graphs, message passing — is developed in Chapter 7, where it is needed.

4.2 The diffusion idea

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021) solve the generative problem with a two-process design inspired by non-equilibrium thermodynamics: a fixed *forward process* gradually corrupts data into pure noise, and a learned *reverse process* undoes the corruption step by step. The forward process used throughout this thesis is the OU channel of Section 3.5: at diffusion time t,

$$x_t \;=\; a\,e^{-t} + \sqrt{\Delta_t}\,\varepsilon, \qquad a \sim p_0, \quad \varepsilon \sim \mathcal{N}(0, I), \qquad \Delta_t = 1 - e^{-2t}, \tag{4.3}$$

so that the noised marginal is the convolution

$$p_t(x) \;=\; \int p_0(a)\,\mathcal{N}\!\Big(x;\, ae^{-t},\, \Delta_t I\Big)\,\mathrm{d}a. \tag{4.4}$$

As t grows, p_t interpolates from the data distribution to $\mathcal{N}(0, I)$. By Anderson's theorem (Section 3.7), the corruption is undone by the reverse SDE (3.15), whose only unknown ingredient is the score $\nabla_x \log p_t(x)$ of the noised marginals. Generative diffusion is therefore, at bottom, *score estimation*: whoever knows $\nabla \log p_t$ for all t can sample from p_0. Note what this design buys relative to Section 4.1: no energy is ever normalised, no equilibrium is ever awaited. The Markov-chain convergence problem of MCMC is replaced by a *fixed-horizon* integration of an SDE whose drift is learned. The partition-function problem has not been solved; it has been *side-stepped*, by working only with objects — scores — from which Z has already been differentiated away.

4.3 Denoising diffusion probabilistic models

The discrete-time formulation (Sohl-Dickstein et al., 2015; Ho et al., 2020) discretises (4.3) into a Markov chain $q(x_s|x_{s-1}) = \mathcal{N}(x_s; \sqrt{1-\beta_s}\, x_{s-1}, \beta_s I)$ with a variance schedule $\beta_1, \ldots, \beta_S$, and learns a reverse chain $p_\theta(x_{s-1}|x_s)$. Training maximises a variational lower bound on the data log-likelihood — the physicist will recognise the Gibbs variational principle (2.11) in modern dress.

Toolbox 4.1 — The evidence lower bound of a diffusion model

For any latent-variable model with joint $p_\theta(x_{0:S})$ and any inference distribution $q(x_{1:S}|x_0)$, Jensen's inequality gives

$$\begin{aligned}\log p_\theta(x_0) &= \log \int p_\theta(x_{0:S})\, \mathrm{d}x_{1:S} = \log \mathbb{E}_q\left[\frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)}\right] \\ &\geq \mathbb{E}_q\left[\log \frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)}\right] \;=:\; \text{ELBO}.\end{aligned}$$

Insert the two Markov factorisations $p_\theta(x_{0:S}) = p(x_S)\prod_s p_\theta(x_{s-1}|x_s)$ and $q(x_{1:S}|x_0) = \prod_s q(x_s|x_{s-1})$, then regroup telescopically using Bayes' rule $q(x_s|x_{s-1}) = q(x_{s-1}|x_s, x_0)\, q(x_s|x_0)/q(x_{s-1}|x_0)$:

$$\begin{aligned}-\text{ELBO} &= D_{\text{KL}}(q(x_S|x_0)\,\|\,p(x_S)) \;-\; \mathbb{E}_q \log p_\theta(x_0|x_1) \\ &\quad + \sum_{s=2}^{S} \mathbb{E}_q\, D_{\text{KL}}(q(x_{s-1}|x_s, x_0)\,\|\,p_\theta(x_{s-1}|x_s))\,,\end{aligned}$$

one denoising step per summand. The decisive point: because the forward chain is Gaussian and conditioned on *both* endpoints, each target $q(x_{s-1}|x_s, x_0)$ is an explicit

Gaussian. Matching it with a Gaussian p_θ reduces each KL term to a weighted regression of the posterior mean — concretely, to predicting the noise ε that was added (Ho et al., 2020). The thermodynamic reading of the ELBO gap as an entropy-production bound goes back to Sohl-Dickstein et al. (2015). □

The practical upshot of Ho et al. (2020) is the loss

$$\mathcal{L}(\theta) = \mathbb{E}_{s,\, a\sim p_0,\, \varepsilon}\Big\|\varepsilon - \varepsilon_\theta\Big(\underbrace{a\, e^{-t_s} + \sqrt{\Delta_{t_s}}\, \varepsilon}_{x_{t_s}},\ s\Big)\Big\|^2, \tag{4.5}$$

a denoising regression, and the empirical discovery that this simplified objective — each noise level weighted equally, all the variational bookkeeping dropped — trains photorealistic image models. Its connection with the score is not incidental, as the next section shows: predicting the noise *is* estimating $\nabla \log p_t$ up to scale.

4.4 Score matching

Suppose we model the score directly by a function $s_\theta(x)$ and wish it to match $\nabla \log p(x)$ for data from p. The naive objective $\mathbb{E}_p \|s_\theta - \nabla \log p\|^2$ seems to require knowing the very score we lack. Two classical results remove the obstruction.

Toolbox 4.2 — Implicit score matching (Hyvärinen)

Expand the square and keep only θ-dependent terms:

$$\mathbb{E}_p \left\|s_\theta - \nabla \log p\right\|^2 = \mathbb{E}_p \left\|s_\theta\right\|^2 - 2\, \mathbb{E}_p \left\langle s_\theta, \nabla \log p\right\rangle + \text{const.}$$

The cross term integrates by parts. Componentwise, using $p\, \partial_i \log p = \partial_i p$ and assuming $p\, s_{\theta,i} \to 0$ at infinity:

$$\int p\ s_{\theta,i}\ \partial_i \log p\ \mathrm{d}x = \int s_{\theta,i}\ \partial_i p\ \mathrm{d}x = -\int p\ \partial_i s_{\theta,i}\ \mathrm{d}x.$$

Therefore

$$\mathbb{E}_p \left\| s_\theta - \nabla \log p \right\|^2 = \mathbb{E}_p \Big[\left\| s_\theta(x) \right\|^2 + 2\, \nabla \cdot s_\theta(x) \Big] + \text{const} :$$

an objective involving only the model and data samples (Hyvärinen, 2005). Elegant, but the divergence term is expensive for large networks — which is why diffusion models use the denoising variant instead. □

Toolbox 4.3 — Denoising score matching (Vincent)

Fix t and consider the noised marginal (4.4), written as $p_t(x) = \int p_0(a)\, q_t(x|a)\, \mathrm{d}a$ with Gaussian kernel $q_t(x|a) = \mathcal{N}(x; ae^{-t}, \Delta_t I)$. Then

$$\nabla_x \log p_t(x) = \frac{\int p_0(a)\, \nabla_x q_t(x|a)\, \mathrm{d}a}{p_t(x)} = \int \underbrace{\frac{p_0(a)\, q_t(x|a)}{p_t(x)}}_{= p(a|x)} \nabla_x \log q_t(x|a)\, \mathrm{d}a,$$

so that

$$\nabla_x \log p_t(x) \;=\; \mathbb{E}\Big[\nabla_x \log q_t(x|a) \,\Big|\, x \Big] :$$

the score of the *marginal* is the posterior average of the score of the *kernel* — and the latter is trivial, $\nabla_x \log q_t(x|a) = -(x - ae^{-t})/\Delta_t$. A standard L^2-projection argument turns this into a regression: for any s_θ,

$$\mathbb{E}_{x \sim p_t} \left\| s_\theta(x) - \nabla \log p_t(x) \right\|^2 = \mathbb{E}_{a,x} \left\| s_\theta(x) - \nabla_x \log q_t(x|a) \right\|^2 + \text{const},$$

because conditional expectation is exactly the L^2 projection onto functions of x (Vincent, 2011). Substituting the Gaussian kernel score and $x = ae^{-t} + \sqrt{\Delta_t}\varepsilon$ gives $\nabla_x \log q_t = -\varepsilon/\sqrt{\Delta_t}$: *learning the score of noisy data is learning to predict the noise*, and the DDPM loss (4.5) is denoising score matching in disguise. □

The continuous-time synthesis (Song and Ermon, 2019; Song et al., 2021) trains one network $s_\theta(x, t)$ against denoising score matching at all noise levels and samples with the reverse SDE (3.15); an accessible exposition is Song (2021), and Karras et al. (2022) systematise the design space. This is the resolution, promised in Section 4.1, of the energy-based model's oldest problem: Song and Ermon replace the energy by its gradient, the equilibrium sampler by a finite-time reverse SDE, and the intractable likelihood by a regression whose targets are exact.

4.5 Flows: the deterministic side of the same coin

Anderson's theorem has a deterministic twin. As noted in Section 3.7, the marginals p_t of the forward SDE are also transported by the noise-free *probability-flow ODE* with velocity $f - \frac{g^2}{2}\nabla \log p_t$ (Song et al., 2021): the same score, consumed by an ODE instead of an SDE, turns generation into a smooth, invertible map from Gaussian noise to data. This places diffusion models inside an older family. *Normalizing flows* (Rezende and Mohamed, 2015; Papamakarios et al., 2021) construct an invertible map T_θ and evaluate likelihoods through the change-of-variables formula; their continuous-time version parametrises the velocity field of an ODE directly (Chen et al., 2018). What diffusion brought to this family is a *simulation-free training signal*: *flow matching* (Lipman et al., 2023) and stochastic interpolants (Albergo and Vanden-Eijnden, 2023) show that one can regress the velocity field of any prescribed interpolation between noise and data — the OU path (4.3) being one choice among many — with the same conditional trick as denoising score matching, never integrating the ODE during training. The unifying statement is worth recording: DDPMs, score SDEs, and flow-matching models all learn a vector field whose ground truth is a *conditional expectation under the noising channel*; they differ in the path of marginals and in whether sampling is stochastic or deterministic. Every exact result of this thesis therefore speaks to the whole family at once: the object we compute in closed form — the conditional expectation behind the field — is the training target of all of them.

4.6 The score as a posterior expectation

The middle line of Vincent's derivation deserves its own section, because it is the bridge on which the whole thesis walks. Rearranging $\nabla \log p_t(x) = \mathbb{E}[-(x - ae^{-t})/\Delta_t \mid x]$:

$$\boxed{\nabla_x \log p_t(x) \;=\; \frac{e^{-t}\,\mathbb{E}[\,a \,|\, x\,] - x}{\Delta_t} \qquad \Longleftrightarrow \qquad \mathbb{E}[\,a \,|\, x\,] = e^{t}\Big(x + \Delta_t\,\nabla_x \log p_t(x)\Big).} \tag{4.6}$$

We refer to (4.6) throughout as the *score–posterior identity*; in the empirical-Bayes literature it is known as Tweedie's formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011). The score of the noisy marginal and the Bayes-optimal denoiser $\mathbb{E}[a|x]$ determine each other exactly, with no assumption whatsoever on p_0. Three readings will recur:

1. *Geometric*: the score points from x towards (the contracted image of) the posterior mean of the clean data — denoising direction and score direction coincide.

2. *Statistical*: estimating the score is solving a Bayesian inference problem — compute a posterior expectation. The score is not *like* an inference target; it *is* one.

3. *Algorithmic*: for sequence data with a Markov prior, that inference problem lives on a tree-shaped graph, so the score is computable by message passing — the programme, made precise in Chapter 7.

For *vector-valued* data $a \in \mathbb{R}^K$ noised coordinate-wise, (4.6) holds componentwise with the *full* posterior:

$$\Big(\nabla_x \log p_t(x)\Big)_k \;=\; \frac{e^{-t}\,\mathbb{E}[\,a_k \,|\, x_0, \ldots, x_{K-1}\,] - x_k}{\Delta_t}. \tag{4.7}$$

The conditioning is on *all* coordinates: even though the noise is independent across frames, the posterior mean of frame k — and hence the k-th score component — depends on every observation, because the prior couples the frames. Equation (4.7) is the precise reason the *joint* score differs from the per-frame marginal score, a distinction that is easy to blur and whose importance for structured data was impressed on this project in supervision;

Chapter 5 quantifies exactly how much coupling the conditioning induces, as a function of t.

4.7 The statistical physics of generative diffusion

A recent line of work analyses diffusion models with the tools of Chapter 2, in the regime where the score is *exact* — either computed from a solvable p_0 or from the empirical measure of n training points. Three findings frame our work.

Regimes and transitions. For high-dimensional solvable data (Gaussian mixtures, spin models), the reverse dynamics crosses sharp dynamical regimes (Biroli and Mézard, 2023; Biroli et al., 2024): a *speciation* time at which the trajectory commits to one mode — a spontaneous symmetry breaking in the sense of Section 2.5 (see also Raya and Ambrogioni, 2023) — and, when the score is learned from finitely many samples, a *memorisation* (collapse) time below which the dynamics condenses onto training points. The phase boundaries are computed with free-energy methods, in direct analogy with the random energy model.

Sampling as physics. Ghio et al. (2024) map generative samplers — flows, diffusions, autoregressive networks — onto statistical-physics problems and delimit where each is efficient, importing the algorithmic-hardness picture of spin glasses (Zdeborová and Krzakala, 2016).

The exactly solvable programme. The common method of these works is to choose p_0 rich enough to be interesting and simple enough that $\nabla \log p_t$ is available in closed form, then study the *exact* generation dynamics. What none of them vary is the *internal structure of a single data point*: their coordinates are exchangeable or i.i.d. conditional on a mode; there is no ordering, no dynamics *inside* the sample.

4.8 Learning as a cascade of phase transitions

The works above put phase transitions on the *sampling* side of generative modelling. A complementary result — Bachtis et al. (2024), which this section examines in detail — puts them on the *training* side, and completes the argument of Section 2.9 that learning itself is a sequence of symmetry breakings. The object of study is the Restricted Boltzmann Machine of Section 2.8, the minimal trainable EBM, with energy $E(v,h) = -v^{\mathsf{T}}Wh - b^{\mathsf{T}}v - c^{\mathsf{T}}h$ and the moment-matching gradient (2.15). The central idea is to track training through the singular value decomposition of the weight matrix,

$$W \;=\; \sum_{\gamma=1}^{r} w_\gamma \, u_\gamma \bar{u}_\gamma^{\mathsf{T}}, \tag{4.8}$$

whose modes $(u_\gamma, \bar{u}_\gamma)$ act as candidate order parameters: the magnetisation of the visible layer along mode γ, $m_\gamma = u_\gamma \cdot v/\sqrt{N_v}$, measures whether the learned Gibbs measure has developed structure in that direction. At initialisation the weights are small and the model is a paramagnet — no direction in configuration space is preferred. The claim of Bachtis et al. (2024) is that training is an *effective cooling*: as the singular values grow, the model crosses a sequence of genuine second-order phase transitions, one for each feature it learns, each signalled by a diverging susceptibility and falling in the mean-field (Curie–Weiss) universality class. The mechanism is visible in a two-line calculation.

Toolbox 4.4 — The retrieval transition of a one-mode RBM

Following Bachtis et al. (2024), take a single Gaussian hidden unit of variance $1/N_v$ coupled to N_v binary visibles through a weight vector w: the energy is $E(v,h) = -h\, w \cdot v + \frac{N_v}{2} h^2$. Integrating out the Gaussian h gives the visible marginal

$$p(v) \;\propto\; \exp\!\left[\tfrac{1}{2N_v}\,(w \cdot v)^2\right]:$$

a Mattis/Hopfield measure (Section 2.7) whose single pattern is w itself, made extensive by the $1/N_v$ variance scaling. Write $w = \bar{w}\,\xi$ with $\xi \in \{-1,+1\}^{N_v}$ and let

$m = \frac{1}{N_v}\sum_i \xi_i v_i$ be the overlap; then $p(v) \propto \exp[\frac{N_v \bar{w}^2}{2} m^2]$. Counting configurations at fixed overlap ($\asymp \exp[N_v H(\frac{1+m}{2})]$ with H the binary entropy) gives the free-entropy density

$$\phi(m) \;=\; \frac{\bar{w}^2}{2}\, m^2 + H\Big(\frac{1+m}{2}\Big) \;=\; \log 2 + \frac{\bar{w}^2 - 1}{2}\, m^2 - \frac{m^4}{12} + O(m^6).$$

This is a Landau expansion: the quadratic coefficient changes sign at $\bar{w}^2 = \|w\|^2/N_v = 1$. Below the threshold the maximiser is $m = 0$ (paramagnet); above it two symmetric maximisers $\pm m^*$ appear continuously (retrieval). The transition is second order with mean-field exponents, and the susceptibility along w diverges as $\|w\|^2/N_v \uparrow 1$. Growing a singular value during training is therefore literally a quench through a Curie–Weiss critical point. □

Why a cascade. With data generated by a Hopfield teacher whose two patterns are correlated combinations $\xi^{1,2} = \eta_1 \pm \eta_2$ of orthogonal directions, the data covariance is rank two with unequal strengths, and the early-time gradient flow of the student's SVD amplitudes along η_1 and η_2 is exponential with *two different rates*. The stronger direction (the "centre of mass" of the clusters) reaches its critical magnitude first: the learned measure splits into two lumps along $\pm\eta_1$. The weaker direction keeps growing at its own rate and crosses criticality later, splitting each lump again — four modes, the full teacher, retrieved in hierarchical order. Each crossing is a distinct second-order transition with its own critical time, its own diverging susceptibility, and hysteresis below it. Learning proceeds coarse-to-fine not by accident of optimisation but because the critical times are ordered by the spectrum of the data.

Evidence on real data. On MNIST, CelebA, and human-genome data, Bachtis et al. (2024) track the susceptibilities of individual SVD modes during training of binary–binary RBMs and observe the same signatures: successive susceptibility peaks that sharpen with system size, finite-size-scaling collapses consistent with mean-field exponents $(\gamma, \nu) = (1, \frac{1}{2})$ at upper critical dimension four, and hysteresis — the operational fingerprints of genuine transitions rather than smooth crossovers. This confirms and sharpens the earlier picture of RBM learning as progressive alignment with the data's principal components (Decelle et al., 2017), of feature emergence as condensation (Tubiana and Monasson, 2017), and of sequential mode learning in linear networks (Saxe et al., 2014); a broad review of the EBM–physics interface is Carbone (2025).

What this means for the thesis. Phase transitions now appear on both faces of generative diffusion: in *sampling*, as speciation events along the reverse trajectory (Biroli et al., 2024), and in *training*, as the cascade just described. Both are hierarchies of symmetry breakings ordered by the data's spectrum. Our research adds the axis that both literatures hold fixed: the structure of the inference problem *inside a single data point*, and how diffusion time t deforms it. The chain we solve is the minimal model in which that question is non-trivial and answerable exactly.

4.9 The gap, restated

Real sequential data — video, audio, trajectories — have exactly the structure the solvable programme leaves out: an internal (frame) time with a Markovian dependence along it. For such data the score (4.7) is a smoothing posterior on a chain, an object with seventy years of inference theory behind it. Yet the questions a diffusion modeller asks about it — how does the coupling structure of $\nabla \log p_t$ deform with t? can it be computed *locally*? what survives a Gaussian parametrisation of the messages when the data are not Gaussian? what does that imply for score architectures? — are not answered by the classical literature, which fixes the noise level, nor by the diffusion literature, which ignores internal structure. Answering them, in the fully solvable setting of the AR(1) chain and its non-Gaussian perturbations, is the contribution of Chapters 5–7.

Chapter 5

The Gaussian Chain Solved Exactly

This chapter answers RQ1. We take the simplest data distribution with genuine internal structure — the stationary Gaussian AR(1) chain of Section 3.2 — corrupt every frame with the OU channel of Section 3.5, and compute the joint score *exactly*, twice: once by linear algebra, once by Bayesian inference. The two routes must agree, and verifying that they do (to machine precision) is the first audit of the computational companion. We then characterise the complete lifecycle of the score's coupling structure — the posterior precision matrix — from the sparse Markov regime at $t \to 0$ to the isotropic regime at $t \to \infty$, including a corrected small-t *band-fill law* and exact closed forms for the bulk of a long chain. Throughout, longer algebraic passages are isolated in toolboxes, and every boxed formula is covered by a named check of the numerical audit (Appendix D).

5.1 The model, and the object of study

Data. A single data point is a trajectory $a = (a_0, \ldots, a_{K-1}) \in \mathbb{R}^K$ of the stationary AR(1) chain (3.5),

$$a_{k+1} = \alpha\, a_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2) \text{ i.i.d.}, \qquad a_0 \sim \mathcal{N}(0, 1), \tag{5.1}$$

with $|\alpha| < 1$ and the *stationary normalisation* $\sigma_\eta^2 = 1 - \alpha^2$, which makes the marginal variance of every frame equal to 1 (Toolbox 3.1). One should think of k as an *internal* (frame) time — the index along the video — to be kept sharply distinct from the diffusion time t.

Corruption. Each frame passes independently through the OU channel (3.11) for a time t:

$$x_k \;=\; e^{-t} a_k + \sqrt{\Delta_t}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1) \text{ i.i.d.}, \qquad \Delta_t = 1 - e^{-2t}. \tag{5.2}$$

Object of study. The *joint score* of the noisy sequence,

$$S(x,t) \;=\; \nabla_x \log P_t(x) \;\in\; \mathbb{R}^K, \qquad P_t(x) = \int_{\mathbb{R}^K} P_0(a) \prod_{k=0}^{K-1} \mathcal{N}\Big(x_k;\, e^{-t} a_k,\, \Delta_t\Big)\,\mathrm{d}a. \tag{5.3}$$

We stress *joint*: the gradient is taken with respect to all K noisy frames simultaneously, under the law of the whole noisy trajectory. The per-frame *marginal* score $\partial_{x_k} \log p_t(x_k)$ is a different, strictly less informative object — for this model it is actually degenerate, as Section 5.3.3 shows — and conflating the two was an instructive early error of this project, corrected in supervision: a sequence is *one* sample from a joint law, not K samples from marginal laws. Only the joint score drives the reverse dynamics of the sequence as a whole, by Anderson's theorem (Section 3.7) applied in $\mathbb{R}^K$.

5.2 The clean joint law: dense covariance, sparse precision

The chain factorisation (3.2) gives the clean joint density

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{K-2} \mathcal{N}\Big(a_{k+1};\, \alpha a_k,\, \sigma_\eta^2\Big) = \mathcal{N}(a;\, 0,\, \Sigma_0), \tag{5.4}$$

a centred Gaussian whose two natural parametrisations — covariance and precision — encode the chain's structure in opposite ways.

Proposition 5.1 (Stationary covariance)**.** $(\Sigma_0)_{ij} = \alpha^{|i-j|}$ *for all* i, j*: a dense, symmetric Toeplitz matrix with geometrically decaying off-diagonals.*

This was derived in Toolbox 3.1 (unrolling the recursion); stationarity is what makes the entries depend on $i - j$ only, i.e. what makes Σ_0 Toeplitz.

Proposition 5.2 (Tridiagonal precision)**.** $Q_0 := \Sigma_0^{-1}$ *is exactly tridiagonal:*

$$(Q_0)_{kk} = \begin{cases} \dfrac{1}{\sigma_\eta^2}, & k \in \{0, K-1\}, \\ \dfrac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < K-1, \end{cases} \qquad (Q_0)_{k,k\pm 1} = -\frac{\alpha}{\sigma_\eta^2}, \tag{5.5}$$

and every entry at distance ≥ 2 *vanishes identically.*

Toolbox 5.1 — Deriving the tridiagonal precision from the log-density

For a centred Gaussian, the precision is the Hessian of the negative log-density. From (5.4),

$$-2\log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{K-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const}, \qquad \sigma_0^2 = 1.$$

Read off the quadratic form term by term. *Off-diagonals:* the only term coupling a_k and a_{k+1} is the transition $(a_{k+1} - \alpha a_k)^2/\sigma_\eta^2$, whose cross coefficient is $-2\alpha/\sigma_\eta^2$, i.e. a precision entry $-\alpha/\sigma_\eta^2$; *no term couples variables at distance* ≥ 2, so those precision entries are exactly zero. *Interior diagonal:* a_k appears in its left transition with coefficient $1/\sigma_\eta^2$ and in its right transition with coefficient α^2/σ_η^2; sum: $(1+\alpha^2)/\sigma_\eta^2$. *Left boundary:* prior term $1/\sigma_0^2 = (1-\alpha^2)/\sigma_\eta^2$ plus right transition α^2/σ_η^2, summing to $1/\sigma_\eta^2$; the right boundary is symmetric (left transition only). □

Remark 5.3 (Tridiagonality is Markovianity, not Gaussianity)*.* A zero in the precision matrix is a conditional independence given all other variables (the Hammersley–Clifford fingerprint, formalised in Section 7.1.1). The derivation above used Gaussianity only to identify the precision with the Hessian of $-\log P_0$; the *support* of that Hessian — nearest neighbours only — comes from the chain factorisation alone. Any Markov chain with smooth transitions has this sparsity pattern in $\nabla^2(-\log P_0)$; the Gaussian case just makes the entries constants. This tridiagonal Q_0 is the only sparse object in the whole story, and every structural result below traces back to it.

The pair (Propositions 5.1, 5.2) is worth contemplating: the *covariance* is dense — every frame correlates with every other, at geometrically decaying strength — while the *precision* is tridiagonal. Correlation is global; conditional dependence is local. The score, we now show, is governed by the second object.

5.3 The noisy joint law and the score, two ways

5.3.1 Route one: linear algebra

Proposition 5.4 (Noised law). $P_t(x) = \mathcal{N}(x;\, 0,\, \Sigma_t)$ *with*

$$\boxed{\Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I_K} \tag{5.6}$$

This is (3.12): the linear channel contracts the signal covariance and adds white noise; Gaussianity is preserved because an affine map plus independent Gaussian noise of a Gaussian is Gaussian. Note the diagonal: $e^{-2t} + \Delta_t = 1$ at every t (variance preservation); only the *off-diagonal* entries, $\alpha^{|i-j|}e^{-2t}$, feel the diffusion time — the noise of different frames is independent, so only the signal correlates.

Since $\log P_t(x) = -\frac{1}{2}x^\top \Sigma_t^{-1} x + \text{const}$,

$$\boxed{S(x,t) \;=\; -\,\Sigma_t^{-1}x \;=\; -\,Q_t\,x, \qquad Q_t := \Sigma_t^{-1}.} \tag{5.7}$$

The joint score is *linear* in x — the hallmark of Gaussianity — and its entire structure is the noisy precision matrix Q_t: every question about the score (which frames talk to which, how strongly, at which t) is a question about a matrix. Sections 5.4 and 5.5 answer those questions completely.

5.3.2 Route two: Bayesian inference

The score–posterior identity (4.7) expresses the same score through the posterior mean of the clean chain:

$$S_k(x,t) = \frac{e^{-t}\,\mathbb{E}[a_k \,|\, x] - x_k}{\Delta_t}. \tag{5.8}$$

The posterior is itself Gaussian, with parameters worth recording once and using forever:

Proposition 5.5 (Posterior of the clean chain)**.** $P(a\,|\,x) = \mathcal{N}(a;\, m,\, J^{-1})$ *with*

$$\boxed{J \;=\; \frac{e^{-2t}}{\Delta_t}\, I \;+\; Q_0, \qquad J\,m \;=\; h := \frac{e^{-t}}{\Delta_t}\, x.} \tag{5.9}$$

Toolbox 5.2 — Posterior precision and field, by completing the square

Bayes' rule with the likelihood (5.2):

$$\begin{aligned} -2\log P(a|x) &= a^{\mathsf{T}} Q_0\, a + \sum_k \frac{(x_k - e^{-t} a_k)^2}{\Delta_t} + \text{const} \\ &= a^{\mathsf{T}}\Big(Q_0 + \tfrac{e^{-2t}}{\Delta_t} I\Big) a - 2\,\tfrac{e^{-t}}{\Delta_t}\, x^{\mathsf{T}} a + \text{const}. \end{aligned}$$

Matching against the generic Gaussian form $a^{\mathsf{T}} J a - 2h^{\mathsf{T}} a$ gives (5.9) directly. Two structural observations. (i) The likelihood is *diagonal* in a: observing each frame through its own channel adds e^{-2t}/Δ_t — the signal-to-noise ratio of the channel, divergent as $1/2t$ at small t, vanishing at large t — to every diagonal entry of the prior precision, and touches nothing else. (ii) Consequently J is *tridiagonal at every* t: the posterior of the clean chain remains a nearest-neighbour MRF; what fills up with t is the precision of the *noisy* law, Q_t, not of the posterior. Keeping J (tridiagonal, posterior) and Q_t (dense, noisy marginal) distinct prevents much confusion. □

Substituting the linear posterior mean $m = J^{-1}h$ into (5.8) must return $-Q_t x$ identically. It does — one can verify the operator identity $\frac{1}{\Delta_t}\Big(e^{-2t} J^{-1} \frac{1}{\Delta_t} - I\Big) = -Q_t$ directly from (5.6) and (5.9) by the Woodbury identity — and the computational companion checks the two routes agree to 2.7×10^{-15} across sweeps of (K, α, t). The agreement is a genuine end-to-end audit: a $K \times K$ inversion against a Bayesian smoothing computation, built from different formulas.

5.3.3 $K = 2$: the smallest example, fully explicit

Everything above, written where nothing can hide. For $K = 2$,

$$\Sigma_t = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \boxed{r = \alpha\, e^{-2t}} \qquad \Longrightarrow \qquad Q_t = \frac{1}{1-r^2} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}, \tag{5.10}$$

$$S_0(x,t) = -\frac{x_0 - r\,x_1}{1-r^2}, \qquad S_1(x,t) = -\frac{x_1 - r\,x_0}{1-r^2}. \tag{5.11}$$

Contrast with the marginal score: each x_k is $\mathcal{N}(0,1)$ at every t (variance preservation), so $\partial_{x_k} \log p_t(x_k) = -x_k$, independent of t and of the other frame — the marginal score of this model *carries no information about the dynamics at all.* The joint score (5.11) instead contains the cross term $-rx_1$: the denoiser of frame 0 borrows evidence from frame 1 with weight $r = \alpha e^{-2t}$ — the whole coupling lifecycle in miniature, equal to α at $t = 0$, decaying exponentially, gone at $t \to \infty$.

5.4 The spectral form and the lifecycle of the precision matrix

How does Q_t interpolate between its two known endpoints — the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$? This section proceeds in the order the question deserves: first the two pieces of linear algebra the analysis rests on, then the *spectral form* of Q_t — the master representation from which everything else follows — and then the study itself: when tridiagonality is lost, how fast, and how much, as diffusion time runs.

5.4.1 The linear algebra we need

Theorem 5.6 (Spectral theorem for symmetric matrices). *Every symmetric $A \in \mathbb{R}^{K\times K}$ admits an orthonormal eigenbasis: there exist an orthogonal matrix U (columns $u_1, \dots, u_K$, $U^\top U = I$) and real numbers $\omega_1, \dots, \omega_K$ with*

$$A \;=\; U\,\mathrm{diag}(\omega_i)\,U^\top \;=\; \sum_{i=1}^{K} \omega_i\, u_i u_i^\top. \tag{5.12}$$

Two consequences carry the whole section. *First*, analytic operations act on the eigenvalues alone: for any function f defined on the spectrum, $f(A) = U\,\mathrm{diag}(f(\omega_i))\,U^\top$; in particular $A^{-1} = U\,\mathrm{diag}(1/\omega_i)\,U^\top$ whenever no eigenvalue vanishes. *Second*, affine images share the eigenbasis: for scalars c, d,

$$(cA + dI)\,u_i \;=\; (c\,\omega_i + d)\,u_i, \tag{5.13}$$

so $cA + dI$ has the *same* eigenvectors as A, with eigenvalues moved by the same affine map. Both facts are elementary; their power here comes from the specific structure of our matrices. The clean covariance Σ_0 is symmetric Toeplitz — constant along diagonals, because the chain is stationary — and the eigenvectors of large symmetric Toeplitz matrices are asymptotically the Fourier modes of the sequence: stationary processes are diagonalised by frequency (Brockwell and Davis, 1991). So the eigenbasis U we are about to fix for all diffusion times is, up to boundary effects, the Fourier basis of the chain.

5.4.2 The spectral form of Σ_t and Q_t

Now apply (5.13) to the forward map (5.6), $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$, which is affine in Σ_0 with $c = e^{-2t}$, $d = \Delta_t$:

Theorem 5.7 (Spectral form). *Let $\Sigma_0 = U\,\mathrm{diag}(\omega_i)U^\top$. Then for* every $t \ge 0$,

$$\boxed{\Sigma_t = U\,\mathrm{diag}\Big(\lambda_i(t)\Big)U^\top, \qquad Q_t = U\,\mathrm{diag}\Big(\frac{1}{\lambda_i(t)}\Big)U^\top, \qquad \lambda_i(t) = e^{-2t}\omega_i + \Delta_t.} \tag{5.14}$$

The content of (5.14) is that diffusion never rotates the problem: *the eigenvectors of Σ_t and Q_t are those of Σ_0, frozen for all t* — only the eigenvalues flow, each along the same affine path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$, all monotonically towards 1 (Figure 5.1). In the eigenbasis the denoising problem is K independent scalar problems at every diffusion time. Two computational corollaries, recorded for later use: factoring Δ_t out of (5.6) gives the *SNR form*

$$Q_t = \tfrac{1}{\Delta_t}\Big(I + \gamma_t\,\Sigma_0\Big)^{-1}, \qquad \gamma_t := e^{-2t}/\Delta_t, \tag{5.15}$$

and multiplying (5.6) by Q_0 and inverting gives the *resolvent form*

$$Q_t = e^{2t}\,Q_0\Big(I + c_t Q_0\Big)^{-1} = e^{2t}\sum_{m\ge 0}(-c_t)^m\,Q_0^{m+1}, \qquad c_t := e^{2t}-1, \tag{5.16}$$

whose Neumann series converges for $\|c_t Q_0\| < 1$, i.e. at small t — exactly where we deploy it below.

5.4.3 Losing tridiagonality: when, how, and how much

The spectral form seems to say that nothing ever changes structurally: the same modes, drifting eigenvalues. The tension with the $t = 0$ picture — an exactly tridiagonal Q_0 — dissolves once one remembers that *sparsity is a property of a matrix in a basis.* In the eigenbasis the problem is diagonal at all times; in the *frame* basis — the operationally meaningful one, since a score architecture reads frames, not Fourier modes — the zeros of Q_0 are a delicate cancellation, and diffusion destroys it. The destruction is precisely quantifiable, and it answers the three questions Jerome's programme poses: *when* is tridiagonality lost, *how* does it happen, and *how much* structure survives at each t.

When: immediately. By (5.14) every entry of Q_t is an analytic function of t. The distance-d entries, identically zero at $t = 0$ for $d \ge 2$, become non-zero for every $t > 0$: the Markov conditional-independence structure is not eroded gradually — it is lost *instantly*, in the sense that no exact zero survives any positive diffusion time.

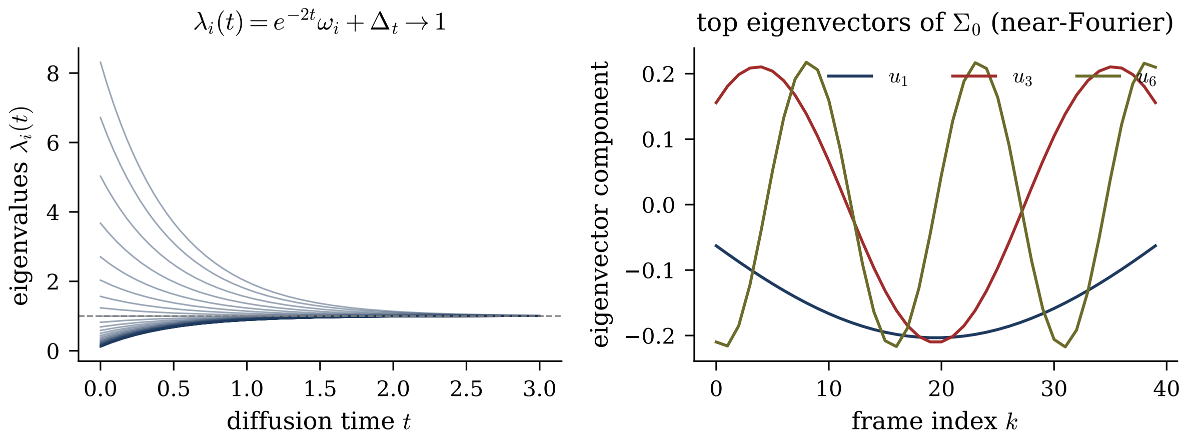

Figure 5.1: Spectral lifecycle. Left: the eigenvalues of Σ_t flow as $\lambda_i(t) = e^{-2t}\omega_i + \Delta_t$, all monotonically towards 1. Right: eigenvectors of Σ_0 — shared by Σ_t and Q_t at every t by Theorem 5.7 — are near-Fourier modes of the chain.

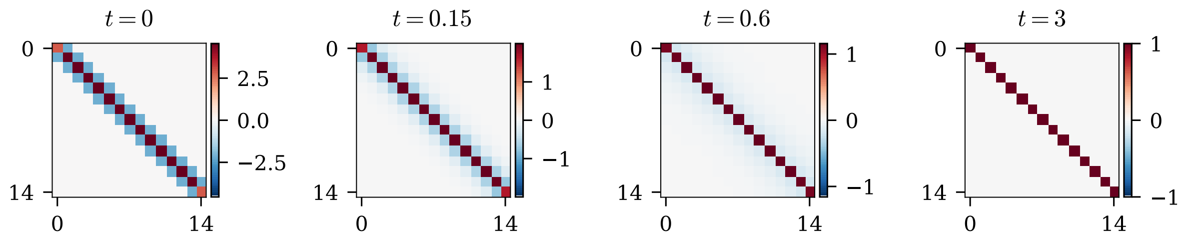

Figure 5.2: Lifecycle of Q_t in the frame basis ($K = 15$, $\alpha = 0.8$): exactly tridiagonal at $t = 0$ (Markov structure), the band fills at intermediate t (every frame couples to every other), and the matrix collapses to the identity as $t \to \infty$ (isotropic noise wins).

How: one diagonal per order in t. The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on (Figure 5.2).

Theorem 5.8 (Band-fill leading order)**.** *For small t and frame distance $d \geq 1$,*

$$\boxed{(Q_t)_{i,i+d} \;=\; (-1)^{d-1}\,(2t)^{d-1}\left(Q_0^d\right)_{i,i+d} \;+\; O(t^d).} \tag{5.17}$$

Toolbox 5.3 — Band fill from the Neumann series

In the resolvent form (5.16), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = \Big(1 + O(t)\Big)\Big[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \cdots\Big].$$

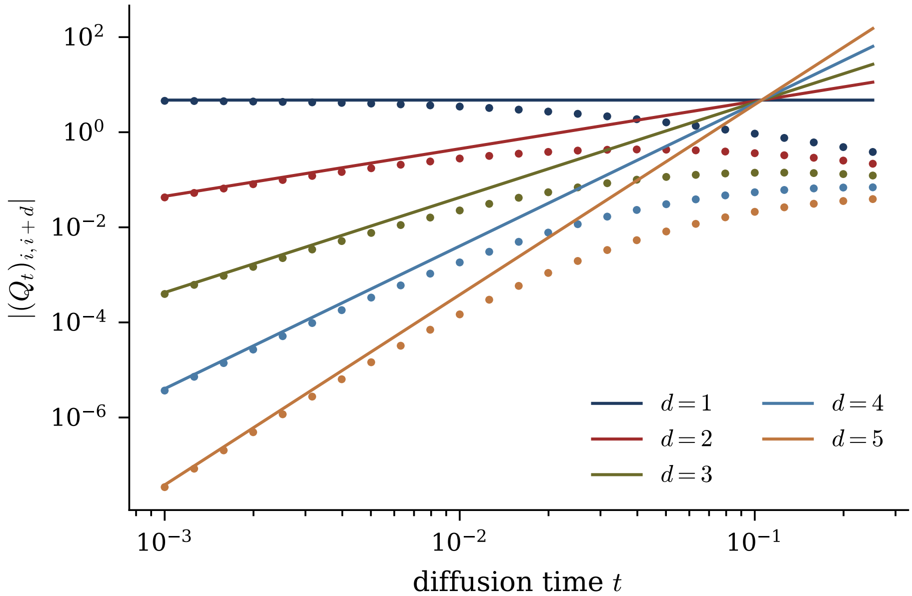

Figure 5.3: Band-fill law, log–log. Measured $|(Q_t)_{i,i+d}|$ against t (dots) and the leading order (5.17) (solid lines — no fitted parameters). The slopes are $d-1 = 0, \ldots, 4$; agreement at small t is at the percent level for every d.

Q_0 is tridiagonal, so Q_0^m has bandwidth m: the entry at distance d receives its *first* contribution from the single term $m = d$ of the series, which carries the prefactor $(-c_t)^{d-1} = (-1)^{d-1}(2t)^{d-1}(1+O(t))$. Every later term is $O(t^d)$. Each additional frame of coupling range costs one power of t: the band fills *outward, one diagonal per order.*

□

Remark 5.9 (Correction of an earlier formula). An earlier working note of this project stated (5.17) with an extraneous factor $1/(d-1)!$. The factor does not follow from the resolvent expansion — the leading distance-d term is a *single* Neumann term, with no combinatorial sum behind it — and the numerical audit rejects it: the no-factorial coefficient matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \ldots, 5\}$ (at $\alpha = 0.9$, $K = 25$), while the factorial version fails by 86% at $d = 3$, 290% at $d = 4$ and over 800% at $d = 5$. Instructively, the error had survived because empirical *slope* fits (exponent $d-1$ in t) are blind to constant prefactors; only auditing the coefficient itself exposed it. Every prefactor in this thesis has since been audited, not only every exponent.

How much: the off-tridiagonal mass across diffusion time. The two regimes join into a single quantitative curve. Define the off-tridiagonal Frobenius mass of Q_t — the norm of everything outside the band that Markov structure would allow. At small t the band-fill law says each new diagonal contributes at its own power of t, so the mass rises from zero, dominated by the $d = 2$ diagonal, linearly in t. At large t, writing $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ and expanding the inverse,

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \tag{5.18}$$

so $\|Q_t - I\|_F \simeq e^{-2t}\|\Sigma_0 - I\|_F$: *all* structure — off-tridiagonal and tridiagonal alike — dies at the universal rate e^{-2t}, and the score collapses onto the isotropic $-x$. In between, the off-tridiagonal mass peaks (Figure 5.4): the noisy sequence is *most* non-Markov at intermediate diffusion times, which is precisely the regime where generative sampling spends its structurally decisive steps. The full lifecycle, in one line: *sparse* (tridiagonal, Markov) at $t = 0$; densifying at rate t^{d-1} per diagonal d; maximally coupled at intermediate t; *isotropic* at $t \to \infty$ at rate e^{-2t}.

5.5 The bulk of a long chain, in closed form

For a long chain, the interior ("bulk") is translation invariant and everything can be solved exactly. By Proposition 5.5 the posterior precision in the bulk is the tridiagonal Toeplitz matrix with

$$\boxed{J_d \;=\; \frac{e^{-2t}}{\Delta_t} + \frac{1+\alpha^2}{\sigma_\eta^2}, \qquad \beta \;=\; -\frac{\alpha}{\sigma_\eta^2}, \qquad \sigma_\eta^2 = 1 - \alpha^2.} \tag{5.19}$$

The diagonal J_d has two parts with opposite behaviour in t: the *evidence* precision e^{-2t}/Δ_t (divergent as $1/2t$ at small t, vanishing at large t) and the constant *prior* precision. The coupling β is pure prior. The single dimensionless ratio $|\beta|/J_d \in (0, \frac{1}{2})$ controls everything below.

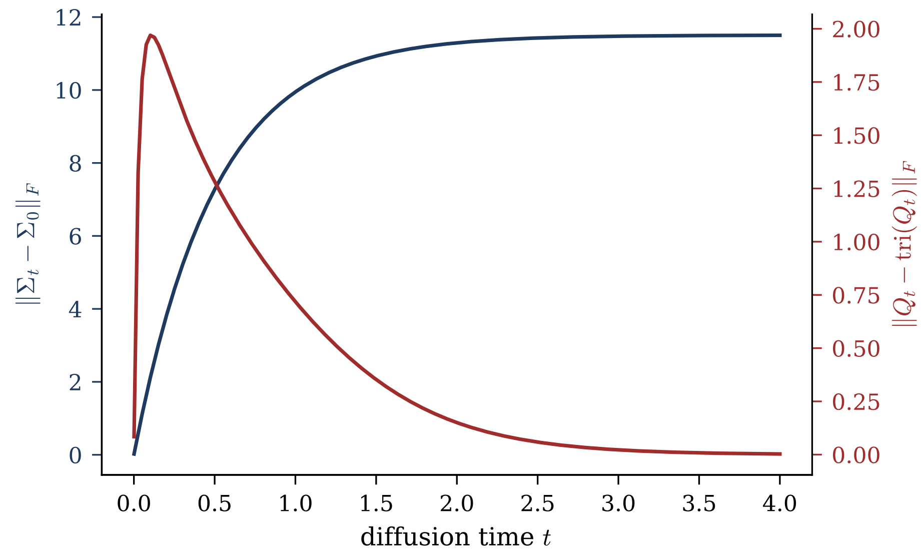

Figure 5.4: Structure lost and recovered, one curve each. $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to the identity; the off-tridiagonal Frobenius mass of Q_t (red) rises from zero, peaks at intermediate t, and decays back to zero: the band fills, then empties.

Theorem 5.10 (Exact bulk variance and geometric covariance)**.** *In the bulk of a long chain, the posterior variance and covariance are*

$$\boxed{V = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \qquad \left(J^{-1}\right)_{i,\,i+d} = q^{\,d}\,V, \qquad q = \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2\,|\beta|} \in (0,1).} \tag{5.20}$$

Toolbox 5.4 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix

Seek the inverse row in geometric form, $(J^{-1})_{i,i+d} = Cq^{\,d}$ for $d \geq 0$ (symmetric in $\pm d$). The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta\, Cq^{\,d-1} + J_d\, Cq^{\,d} + \beta\, Cq^{\,d+1} = 0 \quad \Longleftrightarrow \quad \beta q^2 + J_d\, q + \beta = 0,$$

a quadratic whose root inside the unit disc is $q = (J_d - \sqrt{J_d^2 - 4\beta^2})/(2|\beta|)$ (for $\alpha > 0$, i.e. $\beta < 0$, all inverse entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus). The $d = 0$ relation fixes the prefactor: $J_d C + 2\beta C q = 1$, and since

$2\beta q = -(J_d - \sqrt{J_d^2 - 4\beta^2})$, $C = 1/\sqrt{J_d^2 - 4\beta^2} = V$. Existence of the real square root requires $J_d > 2|\beta|$, which *always* holds here:

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1 - |\alpha|)^2}{\sigma_\eta^2} \; > \; 0$$

for every $|\alpha| < 1$ and $t > 0$ (the AM–GM inequality, with equality only at the excluded $|\alpha| = 1$). This positivity margin returns as a theorem about belief propagation in Chapter 7. □

The number $q(\alpha, t)$ is the *memory of the posterior*: evidence travels $\xi = 1/\log(1/q)$ frames along the chain before fading. Its limits are exactly right:

$$\begin{aligned} t \to 0: &\quad q \approx \frac{|\beta|}{J_d} \to 0 \quad \text{(each frame pinned by its own data);} \\ t \to \infty: &\quad q \to |\alpha| \qquad\quad \text{(posterior} \to \text{prior),} \end{aligned} \tag{5.21}$$

the second limit because $q \to \big((1 + \alpha^2) - (1 - \alpha^2)\big)/(2|\alpha|) = |\alpha|$ when the evidence term dies. As diffusion time runs, the score's internal correlation length climbs monotonically from 0 to the prior's own $1/\log(1/|\alpha|)$.

5.5.1 The locality law

The bulk formulas answer RQ4's quantitative half exactly. Define the *local estimator of radius* r: the posterior mean of a_k given only the observations within distance r, $m_k^{(r)} = \mathbb{E}[a_k \mid x_{k-r}, \ldots, x_{k+r}]$, all other frames marginalised out — the score a *banded* architecture of receptive field r could at best compute. Two facts make the comparison clean: any contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is exact *on its window* — the only approximation is the truncation of range; and both estimators are linear in x, so the error $m_k - m_k^{(r)} = e^\top x$ has a closed-form RMS over the data law, $\mathrm{RMS}_r = \sqrt{e^\top \Sigma_t e}$ — a deterministic number, no sampling involved.

Theorem 5.11 (Locality error decays exactly as q^r)**.** *In the bulk,* $\log \mathrm{RMS}_r$ *decreases*

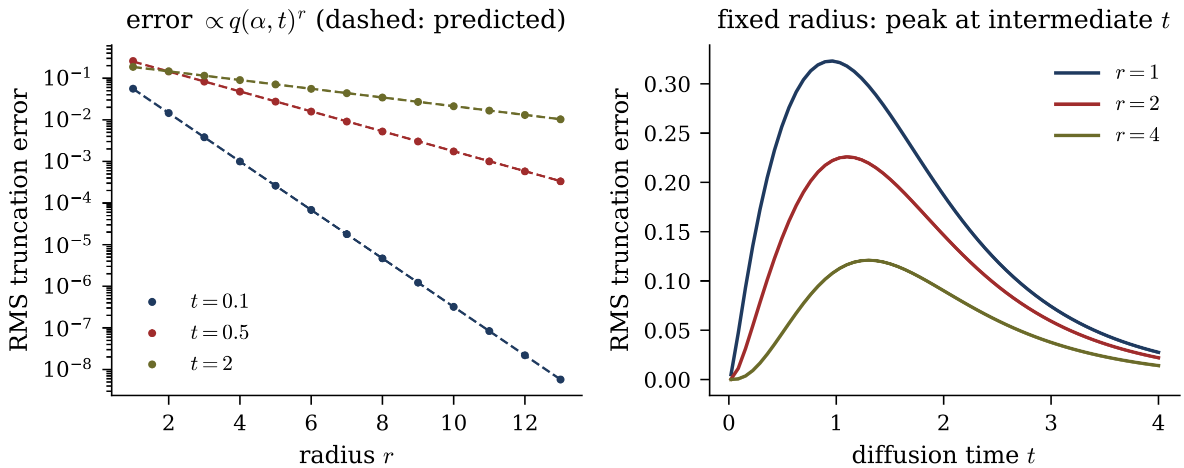

Figure 5.5: Locality of the joint score. Left: exact RMS truncation error of the radius-r local estimator (dots) at three diffusion times; the dashed lines have the *predicted* slopes $q(\alpha, t)^r$ from (5.20) — no fitting. Right: the same error at fixed radius as a function of t: it vanishes at both ends (at $t \to 0$ each frame is pinned by its own observation; at $t \to \infty$ the score itself decouples) and peaks at intermediate t.

linearly in r with slope $\log q(\alpha, t)$*: truncating the evidence range at radius r costs a factor q per discarded frame.*

The mechanism is Theorem 5.10: the influence of evidence at distance d on the posterior mean at k is proportional to $(J^{-1})_{k,k\pm d} \propto q^d$, and the leading discarded evidence sits at distance $r+1$. The audit verifies the *rate*: fitted slopes of $\log \mathrm{RMS}_r$ over $r = 2, \ldots, 13$ at the centre of a $K = 121$ chain match $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ ($\alpha = 0.8$); see Figure 5.5. Note the earlier lesson repeated: single-realisation maximum-error fits are noisy and had misled a preliminary check at large t; the deterministic RMS is the correct instrument.

For a learned score architecture the reading is direct: a receptive field $r \gtrsim \xi \log(1/\varepsilon)$, with $\xi = 1/\log(1/q)$, guarantees relative error ε — and the requirement is most demanding at intermediate diffusion times, exactly where generative sampling spends its critical steps. In the Gaussian benchmark, the cost of locality is not a constant: it is a *rate*, and the rate is the single dial $q(\alpha, t)$.

5.6 Summary of the chapter

For the stationary AR(1) chain under the OU channel: the joint score is $-Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$, equal by the score–posterior identity to the posterior-mean form (5.8) (both audited to machine precision); the clean precision is tridiagonal (Markov), the posterior precision stays tridiagonal at every t, the noisy precision fills at rate t^{d-1} per diagonal (no factorial) and returns to the identity at rate e^{-2t}; the bulk posterior is solved exactly by (5.19)–(5.20), and the single parameter $q(\alpha, t) \in (0, |\alpha|]$ governs simultaneously the posterior correlation length and the exact decay rate of local-truncation error. What this chapter did *not* yet provide is an algorithm: computing $Q_t x$ costs $O(K^3)$ (or $O(K^2)$ with structure), and it is global. Chapter 7 rebuilds the same score from local message passing — whose cost collapses to $O(K)$ scalar updates precisely because, in the Gaussian case, the messages close in a two-parameter family — after Chapter 6 shows what changes when Gaussianity, and with it linearity, is abandoned.

Chapter 6

Beyond Gaussianity: Laplace Innovations

Every result of Chapter 5 rests on one algebraic miracle: Gaussians are closed under the two operations the model performs — multiplying densities and marginalising. This chapter removes the miracle in the most controlled way available: we keep the AR(1) chain, the OU channel, and the entire factor-graph structure, and change *only* the innovation law from Gaussian to Laplace. Heavy-tailed, kinked at the origin, yet still log-concave and analytically friendly, the Laplace distribution is the minimal step outside the Gaussian world — "Laplace and Gaussian, it's not super different" was the supervision verdict, which is precisely why the comparison is informative: every deviation we find is attributable to non-Gaussianity alone, not to a change of structure. We proceed in the order of the research programme agreed in supervision: first the $K = 1$ building block, where the score first becomes nonlinear; then the chain $K \geq 2$, where we exhibit the Markov-structured form of the score, a clean closed form for the boundary message, and the precise point where closed forms die.

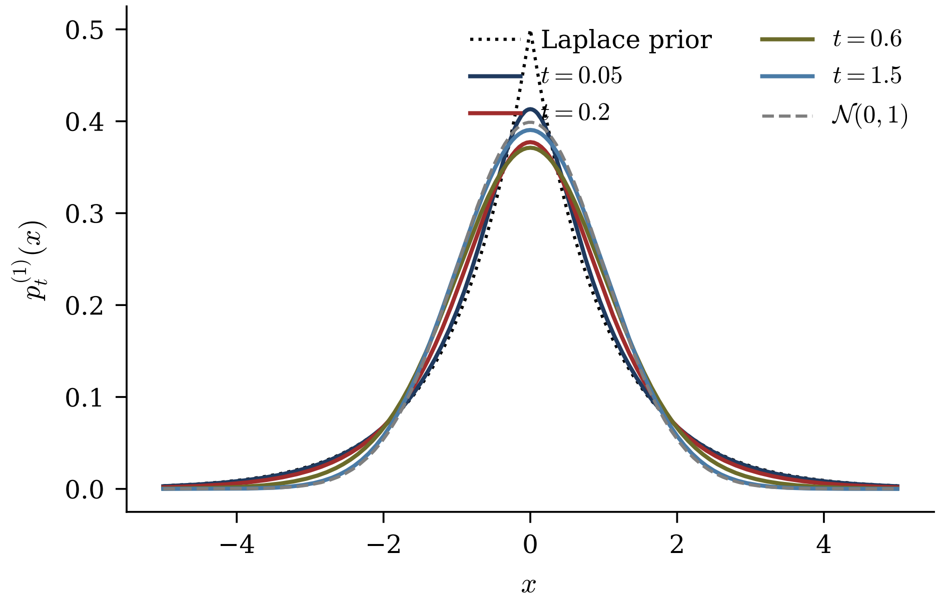

Figure 6.1: Laplace $K = 1$ noised marginal ($b = 1$) at increasing diffusion times, from the kinked Laplace prior (dotted) to the Gaussian attractor (dashed).

6.1 The $K = 1$ warm-up: one frame, no chain

For $K = 1$ there is no Markov structure; the model isolates the single ingredient "non-Gaussian prior meets Gaussian channel". Let

$$a \sim \text{Laplace}(0, b), \qquad p_0(a) = \frac{1}{2b}\, e^{-|a|/b}, \qquad \text{Var}(a) = 2b^2, \tag{6.1}$$

and corrupt through the OU channel: $x = \mu a + \sqrt{\Delta_t}\,\xi$ with $\mu := e^{-t}$, $\xi \sim \mathcal{N}(0, 1)$. The noised marginal is the convolution

$$p_t^{(1)}(x) = \int p_0(a)\, \mathcal{N}\Big(x;\, \mu a,\, \Delta_t\Big)\, \mathrm{d}a. \tag{6.2}$$

Its limits are immediate: as $t \to 0$ the Gaussian kernel becomes a Dirac mass and $p_t^{(1)} \to p_0$ (the kinked Laplace shape); as $t \to \infty$ the kernel loses its a-dependence and $p_t^{(1)} \to \mathcal{N}(0, 1)$, the OU attractor. The family interpolates monotonically between the two (Figure 6.1).

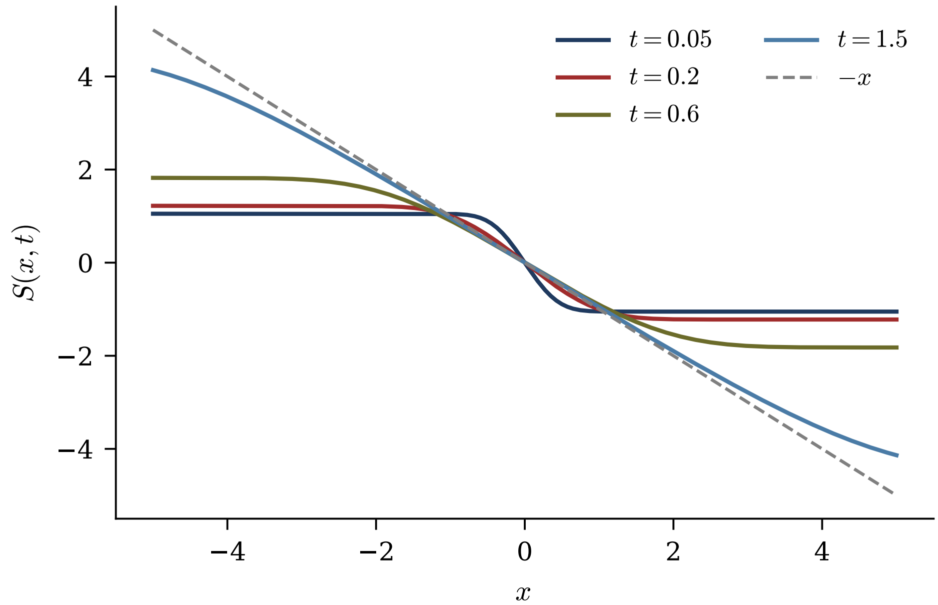

Figure 6.2: Laplace $K = 1$ **score** at the same diffusion times as Figure 6.1; the dashed line is the Gaussian attractor $-x$. The nonlinearity lives in the kink near $x = 0$ and flattens as t grows.

6.1.1 The score becomes nonlinear

The score–posterior identity (4.6) holds for *any* prior: $S(x,t) = (\mu\,\mathbb{E}[a|x] - x)/\Delta_t$. For a Gaussian prior the conditional mean is affine in x and the score is affine; for the Laplace prior $\mathbb{E}[a|x]$ interpolates nonlinearly between the regimes, and the score inherits the nonlinearity. Its two limits are again exact (Figure 6.3): for every fixed $x \neq 0$,

$$t \to 0: \quad S(x,t) \longrightarrow \partial_x \log p_0(x) = -\frac{\operatorname{sign}(x)}{b}, \qquad\qquad t \to \infty: \quad S(x,t) \longrightarrow -x, \tag{6.3}$$

that is: at small t the score converges pointwise to the *sign-shaped* score of the Laplace prior itself, with all the nonlinearity concentrated in the kink near $x = 0$; at large t it flattens into the linear Gaussian-attractor score $-x$. The audit verifies the posterior-expectation route against central finite differences of $\log p_t^{(1)}$ computed by quadrature: maximum disagreement 1.19×10^{-7} over a grid of (b, t, x), at the truncation floor of the finite difference; the Gaussian-attractor limit at $t = 6$ holds to 7.4×10^{-6}.

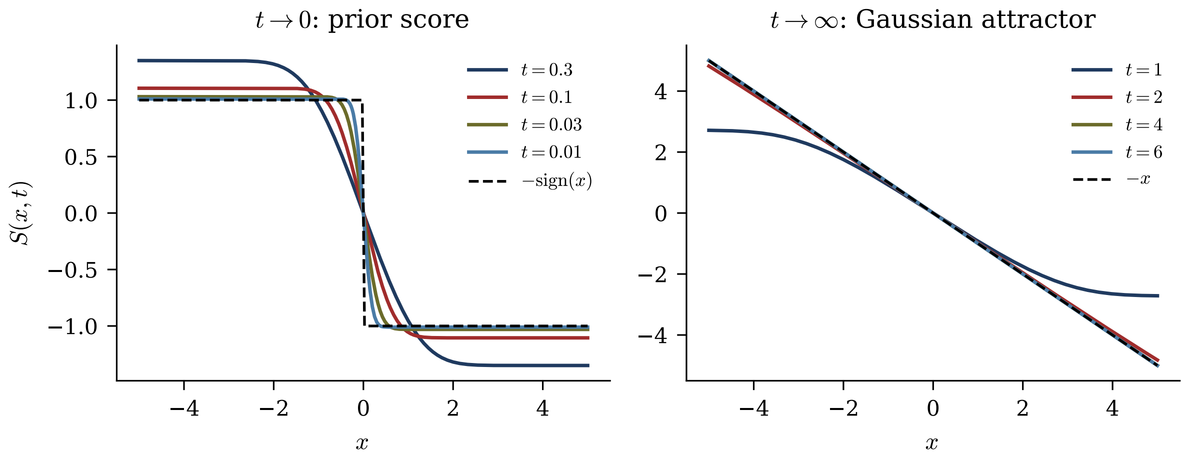

Figure 6.3: Limits of the Laplace $K = 1$ score ($b = 1$). Left: rescaled score for decreasing t, converging pointwise to $-\operatorname{sign}(x)$. Right: unscaled score for increasing t, converging to $-x$.

6.1.2 The Fourier representation: convolution becomes product

The channel writes x as a sum of *independent* variables (μa and $\sqrt{\Delta_t}\xi$), so the density (6.2) is a convolution — awkward in real space, trivial in Fourier space.

Proposition 6.1 (Characteristic-function factorisation)**.**

$$\widehat{p}_t^{(1)}(k) = \underbrace{\frac{1}{1 + b^2\mu^2k^2}}_{c.f.\ of\ \mu a} \cdot \underbrace{e^{-\Delta_t k^2/2}}_{c.f.\ of\ \sqrt{\Delta_t}\,\xi} . \tag{6.4}$$

The proof is the convolution theorem plus two standard transforms: the Laplace characteristic function $(1 + b^2k^2)^{-1}$ (a direct integral), rescaled by μ, and the Gaussian one. What the factorisation buys is a *differential* characterisation of the noised marginal:

Theorem 6.2 (Screened Poisson equation)**.** *The Laplace noised marginal solves the linear, constant-coefficient ODE*

$$\boxed{\left(1 - b^2\mu^2\,\partial_x^2\right) p_t^{(1)}(x) \;=\; G_{\Delta_t}(x)} \qquad G_{\Delta_t}(x) = \frac{e^{-x^2/2\Delta_t}}{\sqrt{2\pi\Delta_t}}. \tag{6.5}$$

Toolbox 6.1 — From the characteristic function to the screened Poisson equation

Multiply (6.4) by $(1 + b^2\mu^2k^2)$ to clear the rational factor:

$$\left(1 + b^2\mu^2k^2\right)\widehat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2},$$

whose right-hand side is exactly the characteristic function of G_{Δ_t}. Now translate back to real space with the differentiation rule: $\widehat{\partial_x f}(k) = ik\,\widehat{f}(k)$, applied twice, gives $\widehat{\partial_x^2 f}(k) = -k^2\widehat{f}(k)$ — multiplication by k^2 in Fourier is $-\partial_x^2$ in real space, and this single minus sign converts $(1 + b^2\mu^2k^2)$ into the operator $(1 - b^2\mu^2\partial_x^2)$. Equation (6.5) follows by inverse-transforming term by term. The audit checks (6.4) against direct quadrature of $\int p_t^{(1)}(x)e^{ikx}\mathrm{d}x$ to 2.9×10^{-6}, the floor of oscillatory quadrature. $\square$

Equation (6.5) is the structural heart of the Laplace case: the noised marginal is the solution of a screened Poisson equation with a Gaussian source, with screening length $b\mu = be^{-t}$ — the memory of the prior's tails, shrinking with diffusion time. It also quantifies the supervision remark that Laplace "is not super different" from Gaussian: the deviation from Gaussianity is governed by a single local differential operator whose strength dies as e^{-t}.

6.1.3 The curvature becomes a field

Differentiating the score–posterior identity once more,

$$H_t(x) := -\partial_x^2 \log p_t^{(1)}(x) = \frac{1}{\Delta_t}\left(1 - \frac{\mu^2}{\Delta_t}\,\mathrm{Var}[a \mid x]\right). \tag{6.6}$$

For a Gaussian prior the posterior variance is x-independent, so H_t is a constant — the score is affine, its Hessian flat. For the Laplace prior $\mathrm{Var}[a|x]$ depends on x, and $H_t(x)$ becomes a nontrivial *field*: the breakdown of Hessian constancy is the second-order shadow of the breakdown of score linearity, and both flow from the same source — a non-Gaussian prior makes $\mathbb{E}[a|x]$ nonlinear. This observation matters for Chapter 7: a Gaussian message is exactly the object that cannot represent an x-dependent curvature, so (6.6) already tells us *what* the Gaussian closure will get wrong, before we measure *how much*.

6.2 The Laplace chain, $K \geq 2$

The Laplace AR(1) chain replaces the Gaussian innovations of (5.1) with Laplace ones:

$$P_0(a) = \frac{1}{(2b)^K} \exp\left(-\frac{1}{b} \sum_{k=0}^{K-1} \left| a_k - \alpha a_{k-1} \right| \right), \qquad a_{-1} := 0. \tag{6.7}$$

The likelihood still factorises across frames, so the posterior inherits the *same* tree-shaped factor graph as the Gaussian model (a fact made precise in Chapter 7): belief propagation remains exact in principle. What is lost is representability — the messages are no longer Gaussian, and computing them requires quadrature or an analytic representation. Three structural results, corresponding to the three threads of the supervision programme, delimit exactly what survives.

6.2.1 The Markov-structured form of the joint score

Theorem 6.3 (Joint score from chain messages)**.** *For* any *chain prior — in particular* (6.7) *— and the OU likelihood, the joint score is, componentwise,*

$$S_k(x,t) = \frac{\mu\, \mathbb{E}[a_k|x] - x_k}{\Delta_t}, \qquad \mathbb{E}[a_k|x] = \frac{\displaystyle\int a_k\, m_k(a_k)\, \mathcal{N}(x_k; \mu a_k, \Delta_t)\, b_k(a_k)\, \mathrm{d}a_k}{\displaystyle\int m_k(a_k)\, \mathcal{N}(x_k; \mu a_k, \Delta_t)\, b_k(a_k)\, \mathrm{d}a_k}, \tag{6.8}$$

where m_k and b_k are the forward and backward BP messages into frame k (Chapter 7).

The content of (6.8) — the direct answer to "express the score using the Markov structure" — is a complexity statement: the K-dimensional posterior integral has been replaced by two *one-dimensional* message recursions plus one one-dimensional integral per frame. Computational locality survives non-Gaussianity intact; what does not survive is the collapse of those 1D integrals into closed-form arithmetic, as we now make precise.

6.2.2 The boundary message in closed form, and where closed forms die

Proposition 6.4 (One-step Laplace integral). *For the Laplace transition* $M(a'|a) = \frac{1}{2b}e^{-|a'-\alpha a|/b}$ *and OU evidence* $\mathcal{N}(x'; \mu a', \Delta_t)$*,*

$$\int_{\mathbb{R}} M(a'|a)\, \mathcal{N}(x'; \mu a', \Delta_t)\, \mathrm{d}a' \;=\; p_t^{(1)}\Big(x' - \mu\alpha\, a\Big) : \tag{6.9}$$

the first backward message of the chain is the universal $K = 1$ *noised marginal, evaluated at a shifted argument.*

Toolbox 6.2 — Innovation change of variable, one step

Fix a and substitute the innovation $r := a' - \alpha a$; the transition density becomes the bare Laplace prior at r, and the Gaussian evidence at $a' = \alpha a + r$ becomes $\mathcal{N}(x' - \mu\alpha a;\, \mu r,\, \Delta_t)$. Hence

$$\int \frac{e^{-|r|/b}}{2b}\, \mathcal{N}\Big(x' - \mu\alpha a;\ \mu r,\ \Delta_t\Big)\, \mathrm{d}r \;=\; p_t^{(1)}\Big(x' - \mu\alpha a\Big)$$

by the very definition (6.2). The shift $\mu\alpha a$ is what frame a predicts for the next frame's noisy observation under the AR(1) dynamics; the $K = 1$ analysis of $p_t^{(1)}$ (screened Poisson equation included) recycles wholesale as the chain's boundary building block.

□

Remark 6.5 (Why iteration destroys the closed form). Equation (6.9) works because the integrand contains *exactly one* Laplace kink times *exactly one* Gaussian. Every subsequent BP step, in either direction, violates this: the incoming message already carries the propagated structure of several frames — a sum of products of kinks and Gaussians — so the next integrand contains two or more absolute-value kinks. On each sign-quadrant the integral reduces to half-line Gaussians, but globally it is a sum of error functions whose arguments depend on the conditioning variable non-affinely: the clean form does not survive one iteration. Contrast the Gaussian chain, where every step is Gaussian-times-Gaussian and closure propagates forever. This is the precise, structural location of the difference between the two benchmarks — and the precise reason the Gaussian projection of Chapter 7 is an *approximation* here, with an error to be measured rather than an identity to be proved.

6.2.3 Innovation coordinates: the coupling has to live somewhere

The one-step trick suggests decoupling the whole prior at once. Define the triangular change of variables $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$; its inverse is $a = Lr$ with $L_{ij} = \alpha^{i-j}\mathbf{1}_{i\geq j}$ unit lower-triangular, so the Jacobian is 1 and

$$P_0(r) = \frac{1}{(2b)^K}\prod_{k=0}^{K-1} e^{-|r_k|/b} : \tag{6.10}$$

in innovation coordinates the prior factorises into i.i.d. Laplace variables — as decoupled as it can be. But substituting $a = Lr$ in the likelihood produces

$$p(x|r) \propto \exp\Big(-\tfrac{\mu^2}{2\Delta_t}\, r^\top L^\top L\, r + \tfrac{\mu}{\Delta_t}\, x^\top L\, r\Big), \tag{6.11}$$

and $L^\top L$ is *dense* (its entries sum geometric series in α and vanish nowhere). The AR(1) coupling did not disappear; it migrated from the prior to the likelihood. In the original coordinates the prior carries the chain and the likelihood is diagonal; in innovation coordinates the prior is diagonal and the likelihood carries the chain. There is no basis in which both are diagonal at once — the non-Gaussian counterpart of the Gaussian statement that Q_t is tridiagonal in the frame basis and diagonal in the spectral basis, but never both (Section 5.4).

6.3 What survives, what is open

Chapter results, classified. *Exact*: the noised marginal limits, the score limits (6.3), the characteristic-function factorisation (6.4), the screened Poisson equation (6.5), the Markov-structured score (6.8), the one-step boundary form (6.9), the innovation factorisation (6.10)–(6.11), and the structural breakdown argument of Remark 6.5. *Numerical*: the quadrature audits quoted above (posterior-expectation route vs finite differences; characteristic function vs direct integration). *Open*: a quadrature-free representation of general (non-boundary) messages at $K \geq 2$, e.g. by controlled small-α expansion; and a basis-independent statement of which locality of the joint score survives non-Gaussianity (candidates: sparsity of the x-dependent Hessian in a suitable basis, low-rank corrections to a tridiagonal core). An earlier conjecture of this project — that the Laplace $K = 2$ Hessian is approximately rank-two — proved regime-dependent under quick-resolution tests and is *not* asserted here.

The chapter leaves us exactly where the research programme wants us: the Laplace chain is structurally identical to the Gaussian one — the same tree, the same two-sweep message schedule, the same score–posterior identity — but its messages are genuinely infinite-dimensional functions, so the number of message *updates* stays linear in K while the cost of each update is no longer bounded. The question "what does it cost to force the messages Gaussian?" is now sharply posed; Chapter 7 answers it.

Chapter 7

The Score as Posterior Expectation: Belief Propagation and the Kalman Filter

Chapter 5 computed the joint score by inverting a $K \times K$ matrix; the score–posterior identity (Section 4.6) says the same object is a posterior expectation. This chapter takes the Bayesian route seriously and rebuilds the score by *message passing*: local computations on the graph of the posterior. It answers RQ2 and RQ3, and it is deliberately the most self-contained chapter of the thesis: the graphical-model formalism — Markov random fields, factor graphs, the sum–product algorithm — is developed here, where it is used, and in the depth its central role deserves. The plan: first the structure of the posterior (a tree), then the sum–product algorithm and its exactness, with the honest accounting of what "linear cost" does and does not mean for continuous variables; then the Gaussian case, where the messages close algebraically — no central-limit argument anywhere — and the sweeps turn out to be, update by update, the classical Kalman filter and smoother; finally the Gaussian closure on genuinely non-Gaussian chains, measured against exact grid quadrature. The per-node closures of statistical physics (AMP, the TAP equations), their exact breakdown time on the chain, and the critical coupling $\alpha_c = \sqrt{2} - 1$ are developed in Appendix C: the results are complete and audited, but they sit off the main line of the argument.

7.1 The posterior and its graph

Fix the setting of Chapters 5–6: a Markov chain prior $P_0(a) = p_0(a_0) \prod_k M(a_{k+1}|a_k)$, per-frame OU evidence, and the posterior

$$p(a \,|\, x) \;\propto\; \underbrace{\psi_{\rm pr}(a_0) \prod_{k=0}^{K-2} \psi_{\rm tr}^{(k)}(a_k, a_{k+1})}_{\text{chain prior}} \; \underbrace{\prod_{k=0}^{K-1} \psi_{\rm ob}^{(k)}(a_k)}_{\text{evidence}}, \qquad \psi_{\rm ob}^{(k)}(a_k) = \mathcal{N}\Big(x_k;\, e^{-t} a_k,\, \Delta_t\Big). \tag{7.1}$$

Structurally this is a *hidden Markov model* (Rabiner, 1989): a latent chain observed through independent per-frame noise, and by the score–posterior identity the k-th score component needs exactly the smoothing marginal $\mathbb{E}[a_k|x]$ of (7.1). Before computing, we make the structure of this object precise, in the language built for it.

7.1.1 Markov random fields

Definition 7.1 (Markov random field)**.** Let $G = (V, \mathcal{E})$ be an undirected graph with one random variable x_i per node. The collection $x = (x_i)_{i \in V}$ is a *Markov random field* (MRF) if it satisfies the local Markov property: for every node i,

$$p\Big(x_i \,\Big|\, x_{V \setminus \{i\}}\Big) \;=\; p\Big(x_i \,\Big|\, x_{\partial i}\Big), \tag{7.2}$$

where ∂i is the set of neighbours of i in G.

The fundamental structure theorem connects this conditional-independence definition to the energy picture of Chapter 4: by the Hammersley–Clifford theorem (proved in Besag, 1974), a strictly positive density is an MRF on G if and only if it factorises over the *cliques* of G,

$$p(x) \;=\; \frac{1}{Z} \prod_{C \in \text{cliques}(G)} \psi_C(x_C) \;=\; \frac{1}{Z} \exp\Big[-\sum_C E_C(x_C)\Big]. \tag{7.3}$$

Conditional independence, graphical locality, and additive local energies are one and the same thing. The Ising model (2.12) is the textbook example; the decisive example for us is one-dimensional: the chain factorisation (3.2) makes a Markov chain an MRF on a line, with pairwise nearest-neighbour energies. For the Gaussian AR(1) chain each energy term is quadratic, so the coupling matrix of the quadratic form connects only neighbours — this is the Hammersley–Clifford reading of the tridiagonal precision of Chapter 5 (Remark 5.3), now with its proper name. And because the likelihood factors of (7.1) touch one variable each, conditioning on the observations does not enlarge any clique: *the posterior is again an MRF on the line*, whatever the innovation law.

7.1.2 Factor graphs, and the tree

MRFs record *which* variables interact but not *how* the joint factorises (a triangle of pairwise terms and one triple term have the same graph). The representation that makes factorisation itself graphical — and on which message passing is most naturally defined — is the *factor graph* (Kschischang et al., 2001; Koller and Friedman, 2009).

Definition 7.2 (Factor graph)**.** Given a factorisation $p(x) = \frac{1}{Z}\prod_a \psi_a(x_{\partial a})$, its factor graph is the bipartite graph with a *variable node* for each x_i, a *factor node* for each ψ_a, and an edge (i, a) whenever $x_i \in x_{\partial a}$.

The factor graph of (7.1) has K variable nodes, $2K$ factor nodes (1 prior, $K-1$ transitions, K observations — the data x_k are constants clamped inside their factors, not nodes), and $3K - 1$ edges: each transition factor has degree 2, prior and observations degree 1. A connected graph with (nodes $-\,1$) edges is a *tree*. Pictorially it is a *caterpillar*: a spine of variables and transition factors, one evidence leaf per vertebra — in the words of the supervision discussion, "very tree-like: a straight line with things on the side". Every interior variable has degree three; every message update combines a small, fixed number of inputs. Recording the conclusion the rest of the chapter exploits:

Proposition 7.3. *For any Markov-chain prior and any per-frame observation model, the posterior $p(a \mid x)$ factorises over a tree-shaped factor graph, and all K smoothing marginals* *are computed exactly by $O(K)$ message* updates *(one forward and one backward sweep).*

Note carefully what Proposition 7.3 does *not* say: it does not say the total *cost* is $O(K)$, because it does not say the messages are finitely parametrisable. For discrete chains each message is a probability vector and each update is finite arithmetic; for continuous chains each message is a *function*, each update is an integral in function space, and the cost per update is unbounded in general. The count of updates is linear in K; the price of an update depends entirely on how messages are represented. Keeping "tree-exactness of the algorithm" separate from "representability of the messages" is the conceptual hinge of this thesis: the first is a graph property, settled here once and for all; the second is where all the difficulty — and all of Sections 7.3 and 7.4 — lives.

7.2 The sum–product algorithm

On a factor graph, sum–product belief propagation passes messages along every edge (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009): variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages,

$$\nu_{i\to a}(a_i) \propto \prod_{b\in\partial i\setminus a} \hat{\nu}_{b\to i}(a_i), \qquad \hat{\nu}_{a\to i}(a_i) \propto \int \psi_a(a_{\partial a}) \prod_{j\in\partial a\setminus i} \nu_{j\to a}(a_j)\, \mathrm{d}a_{\partial a\setminus i}, \tag{7.4}$$

with the belief at node i proportional to $\prod_{a\in\partial i} \hat{\nu}_{a\to i}(a_i)$. The "exclude the recipient" rule is the heart of the algorithm: the message $i \to a$ summarises everything node i knows *except* what it learned from a, so that no piece of evidence is ever counted twice along a path. On a tree this bookkeeping is perfect: each edge separates the graph into two disjoint halves, the message crossing it summarises exactly the half behind it, and the equations (7.4) have a unique solution, reached in one inward and one outward pass, whose beliefs are the *exact* marginals (Mézard and Montanari, 2009, Theorem 14.1). On graphs with cycles the same updates, iterated, define *loopy* BP — extraordinarily useful, but approximate (Weiss and Freeman, 2001; Malioutov et al., 2006); none of that is needed here, because our graph is a tree.

For *discrete* variables each message is a finite probability vector and (7.4) is arithmetic. For *continuous* variables each message is a probability *density* — an infinite-dimensional object — and the update integrals must be carried out in function space. This is the obstruction, identified sharply in supervision, that makes “BP on continuous chains” a research question rather than a textbook exercise. There are exactly three ways out: (i) discrete data (retreat to finite alphabets); (ii) a family of densities closed under the updates — the Gaussian route, exact for our Gaussian chain, as we prove next; (iii) numerical function representation — the grid route, which Section 7.4 uses as exact reference for non-Gaussian chains.

7.2.1 The chain sweeps, with the bookkeeping made explicit

On the chain, (7.4) collapses into one forward and one backward sweep. The bookkeeping deserves unusual care — an earlier version of this project’s notes fell into a trap here, caught only by the numerical audit.

Proposition 7.4 (Convention A). *Define messages*

$$\mu_{\to 0}(a_0) := \psi_{\mathrm{pr}}(a_0), \quad \mu_{\to k+1}(a_{k+1}) := \int \psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1})\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\to k}(a_k)\, \mathrm{d}a_k, \tag{7.5}$$

$$\mu_{\leftarrow K-1}(a_{K-1}) := 1, \qquad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\mathrm{tr}}^{(k-1)}(a_{k-1}, a_k)\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\leftarrow k}(a_k)\, \mathrm{d}a_k. \tag{7.6}$$

Then for every k,

$$p(a_k \,|\, x) \;\propto\; \mu_{\to k}(a_k)\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\leftarrow k}(a_k) : \tag{7.7}$$

the forward message into a_k carries the evidence $x_0, \ldots, x_{k-1}$ only, the backward message carries $x_{k+1}, \ldots, x_{K-1}$ only, and the local evidence enters once, at combination time.

The proof is direct marginalisation: integrating (7.1) over all a_j, $j \neq k$, splits at a_k by the Markov property into independent left and right halves, and peeling the innermost variable of each half yields the recursions. What deserves the reader’s full attention is the *bookkeeping*:

Remark 7.5 (The double-counting trap). A natural-looking alternative (call it Convention B) absorbs the local evidence into the forward message, $\mu^{\mathrm{B}}_{\to k} = \mu^{\mathrm{A}}_{\to k} \cdot \psi^{(k)}_{\mathrm{ob}}$. Under Convention B the product (7.7) counts x_k *twice* and must be corrected by dividing one factor back out. Convention A is the bookkeeping under which the textbook combination rule is correct *as written*. The audit was decisive: under Convention A the BP score matches the matrix score to 10^{-14}; under naive Convention B the disagreement is order one. This is the single most error-prone point of the construction — hence every message in this thesis is stated with its evidence content made explicit.

Each sweep computes $K - 1$ messages, and the combination at each node multiplies three univariate functions: $O(K)$ updates, as Proposition 7.3 promised — with the cost of each update still an open bill, to be settled family by family.

7.3 The Gaussian case: closure, and the Kalman filter

For the Gaussian chain, two elementary facts make the message space finite-dimensional — and they are the *only* facts needed.

Toolbox 7.1 — The two closure lemmas of Gaussian message passing

Lemma 1 (products). For Gaussians in the same variable,

$$\mathcal{N}(a; m_1, v_1) \times \mathcal{N}(a; m_2, v_2) \;\propto\; \mathcal{N}(a; m, v), \qquad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}:$$

add the exponents; the sum of two quadratics in a is a quadratic in a, with precisions and precision-weighted means adding. (In the natural parameters $\lambda = 1/v$, $\theta = m/v$ the rule is literally $(\lambda, \theta) \mapsto (\lambda_1 + \lambda_2,\, \theta_1 + \theta_2)$.)

Lemma 2 (affine marginalisation). Pushing a Gaussian through the AR(1) transition,

$$\int \mathcal{N}(a'; \alpha a, \sigma_\eta^2)\, \mathcal{N}(a; m, v)\, \mathrm{d}a \;=\; \mathcal{N}(a';\, \alpha m,\, \alpha^2 v + \sigma_\eta^2),$$

because $a' = \alpha a + \eta$ with independent Gaussian summands: means and variances transform as stated, and Gaussianity is preserved. The same computation read in the reverse direction gives the backward analogue $\mathcal{N}\big(a;\, m'/\alpha,\, (v' + \sigma_\eta^2)/\alpha^2\big)$.
Together: the Gaussian family is closed under exactly the two operations the recursions (7.5)–(7.6) perform. Note also that the local evidence, read as a function of a_k, is itself Gaussian — $\psi_{\text{ob}}^{(k)}(a_k) \propto \mathcal{N}(a_k;\, e^t x_k,\, \Delta_t e^{2t})$, a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t, the pinning strength of Chapter 5. □

Theorem 7.6 (Gaussian closure of BP on the chain)**.** *For the Gaussian AR(1) chain under the OU channel, every BP message and every belief is exactly Gaussian, at every node, for every* (K, α, t)*. Each message is carried by two numbers, so each update is* $O(1)$ *arithmetic and the two sweeps compute the exact posterior marginals — hence, via the score–posterior identity, the exact joint score — at total cost* $O(K)$*.* Here, and only here, the linear update count of Proposition 7.3 becomes a linear cost.

Derivation. Induction along each sweep. The initial messages are Gaussian (the prior; the flat message is the $\lambda \to 0$ limit of the family). Each step of (7.5)–(7.6) is one application of Lemma 1 (absorb the local evidence) followed by one of Lemma 2 (push through the transition); the combination (7.7) is Lemma 1 twice. Exactness follows from tree-exactness of sum–product. □

Remark 7.7 (Closure is algebra, not a central limit theorem). The closure holds at node degree two just as it would at degree two thousand: it is a property of the Gaussian *family* under multiplication and affine marginalisation, not an aggregation-of-many-influences effect. This answers precisely the doubt raised in supervision — whether Gaussian messages on a chain could be trusted, given that the usual central-limit justification needs many incoming messages. For *exact* Gaussian BP, no CLT is invoked anywhere, so there is nothing to fail. The CLT enters only to justify the *per-node* closures of statistical physics (AMP/TAP) — and there, on the chain, it fails exactly as predicted; Appendix C makes that failure quantitative, complete with an exact breakdown time and the critical coupling $\alpha_c = \sqrt{2} - 1$.

7.3.1 The sweeps are the Kalman filter and smoother

Executing the Gaussian sweeps in mean–variance arithmetic reproduces, line by line, the most famous algorithm of estimation theory. The identification deserves to be displayed rather than asserted.

Toolbox 7.2 — The Kalman filter, derived from the BP sweep

Write the forward message in Convention A as $\mu_{\to k} = \mathcal{N}(a_k;\, m_{k|k-1},\, P_{k|k-1})$ — in filtering language, the *predicted* state given past observations. One BP step does two things.

Update (absorb the evidence; Lemma 1). Multiplying by the pseudo-observation $\mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$ gives the *filtered* Gaussian $\mathcal{N}(a_k; m_{k|k}, P_{k|k})$ with

$$\frac{1}{P_{k|k}} = \frac{1}{P_{k|k-1}} + \frac{e^{-2t}}{\Delta_t},$$
$$m_{k|k} = P_{k|k}\left(\frac{m_{k|k-1}}{P_{k|k-1}} + \frac{e^{-t}x_k}{\Delta_t}\right) = m_{k|k-1} + G_k\Big(e^t x_k - m_{k|k-1}\Big),$$

where the second form defines the *Kalman gain* $G_k = P_{k|k}\, e^{-2t}/\Delta_t \in (0,1)$: the filtered mean is the prediction, corrected by a gain-weighted innovation — the celebrated predictor–corrector structure (Kalman, 1960).

Predict (push through the dynamics; Lemma 2). Propagating through the AR(1) transition,

$$m_{k+1|k} = \alpha\, m_{k|k}, \qquad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2 :$$

contract the mean, inflate the variance by the innovation noise. The forward BP sweep *is* the Kalman filter — not analogous to it, equal to it, message by message.

Smoothing (combine both directions). The belief (7.7) multiplies the filtered forward Gaussian by the backward message (Lemma 1 once more), producing the posterior marginal given *all* observations, $\mathcal{N}(a_k; m_{k|K-1}, P_{k|K-1})$. Carrying out the same algebra backwards recursion-by-recursion yields the classical fixed-interval smoother of Rauch et al. (1965): the two-sweep BP schedule on the chain is exactly the Kalman filter followed by the RTS smoother, the Gaussian specialisation of sum–product (Kschischang et al., 2001; Bishop, 2006). □

The score–posterior identity then gives the punchline of RQ2: *the k-th component of the joint diffusion score is an affine function of the Kalman-smoothed estimate of frame k*,

$$S_k(x,t) \;=\; \frac{e^{-t}\, m_{k|K-1} - x_k}{\Delta_t}. \tag{7.8}$$

Score computation for Markov sequences is a smoothing problem, and seventy years of smoothing theory apply verbatim.

Machine-precision agreement. The computational companion runs BP against the matrix route of Chapter 5 over 80 configurations — $K \in \{2,3,5,8,16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random x: maximum disagreement 1.3×10^{-15} on posterior means, 3.9×10^{-15} on variances, 1.2×10^{-14} on scores — double-precision rounding accumulated along the recursion, nothing more. An independent implementation, written from scratch against the supervision-call specification alone, reproduces the same agreement; and a worked $K = 3$ instance with every message spelled out numerically is part of the companion notebooks. One qualitative detail of that instance is worth repeating: at $x = (1.2, -0.4, 0.8)$ the score component S_1 is *positive* although $x_1 < 0$ — the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing, the joint effect of Section 5.3.3, in numbers.

7.4 The Gaussian closure beyond Gaussianity

On the Laplace chain of Chapter 6, BP is still exact in principle but its messages are genuinely infinite-dimensional (Remark 6.5). This is where the closure question becomes a measurement. Two solvers are compared on identical data.

Exact reference: grid BP. Messages are represented by their values on M equispaced points in $[-A, A]$ and updated by trapezoidal quadrature — option (iii) of Section 7.2, at cost $O(KM^2)$: the honest price of function-valued messages. Calibration on the exactly solvable Gaussian chain shows the trapezoid rule on analytic, rapidly decaying integrands converges *spectrally* (aliasing error $\sim e^{-2\pi^2\sigma^2/\mathrm{d}x^2}$), so failure is abrupt, not gradual: at $M = 101$, $A = 8$ the score error sits at the 10^{-15} floor down to $t = 0.02$ and fails only when the likelihood width $\sqrt{2t}$ approaches the grid step. The working configuration for all non-Gaussian experiments is $M = 401$, $A = 8$ (reference $M = 801$), with the resolution constraint $\sqrt{2t} \gtrsim 3\,\mathrm{d}x$ checked automatically; grid self-convergence error remains 10^4–10^7 times smaller than the effect being measured at every t. Within its validated regime, grid BP *is* the exact score.

The object under test: Gaussian-projected BP. The same sweeps, with every message forced into the Gaussian family by moment matching — the algorithm one would actually deploy when quadrature is unaffordable, and the chain analogue of the per-node closures of Appendix C.

The error instrument. All score errors are amplified posterior-mean errors: for any two solvers producing means $\widehat{m}, m_{\mathrm{ref}}$ and scores by the score–posterior identity,

$$\widehat{s} - s_{\mathrm{ref}} = \frac{e^{-t}}{\Delta_t}\Big(\widehat{m} - m_{\mathrm{ref}}\Big), \qquad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \ \ (t \downarrow 0), \tag{7.9}$$

an exact identity (algebra, not approximation) whose numerical residual is logged in every experiment and sits at 10^{-15}–10^{-12} throughout. Equation (7.9) explains *a priori* why small diffusion times are the danger zone: whatever the closure does to the posterior mean is amplified by $1/2t$ in the score.

7.4.1 Findings on the Laplace chain

On the Laplace chain ($\rho = 0.85$, $n = 40$, 60 trials per t, reference $M = 801$): the median relative score error of the Gaussian closure is

t	0.02	0.08	0.4	2.4
median relative score error	0.39	0.17	3.5×10^{-2}	7.6×10^{-5}

with failure probability $\mathbb{P}(\mathrm{err} > 0.1)$ equal to 1 for $t \leq 0.08$ and 0 for $t \geq 0.9$ (Figure 7.1). The closure error is thus of order tens of percent in the small-t regime and decays below 10^{-3} once diffusion noise Gaussianises the tilted posteriors — the quantitative form of the supervision intuition that "at long diffusion times the world is Gaussian, so probably you have no problem". Inspection of the worst trials shows the mechanism: the reference belief is sharply non-Gaussian (occasionally bimodal), the single Gaussian commits to a compromise mean, and (7.9) amplifies the shift. The score *direction*, however, remains far more accurate than its magnitude throughout: the median cosine similarity between projected and reference scores is 0.92 even at $t = 0.02$ (where the magnitude error is 39%), exceeds 0.99 for all $t \geq 0.12$, and the worst single trial of the whole sweep stays above 0.78.

7.4.2 Varying the innovation: what makes closure hard

Sweeping six innovation families with *identical* covariance $\rho^{|i-j|}$ — Laplace, Student-t ($\nu = 5, 8$), and symmetric Gaussian mixtures of increasing separation — isolates the shape of the innovation law as the only moving part. Three findings:

1. *Noise level dominates.* Every family's error decays to $\sim 10^{-3}$ or below by $t = 2.4$; all meaningful closure error lives at $t \lesssim 0.4$.

2. *Both signs of non-Gaussianity hurt; bimodality hurts most.* At $t = 0.05$, $\rho = 0.5$: strongly bimodal mixture 0.94, Laplace 0.52, Student-$t_{\nu=5}$ 0.32, mildly bimodal mixture 0.07. A single Gaussian cannot represent two posterior modes, and commits to the gap between them.

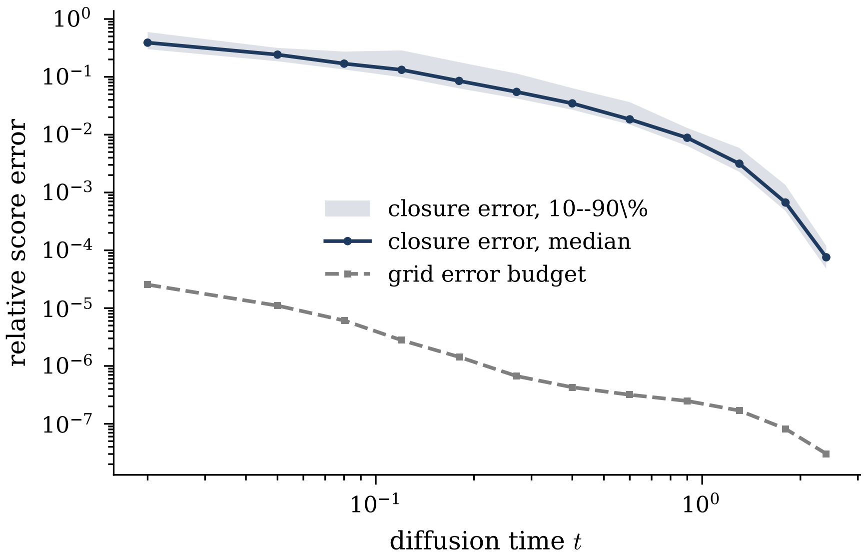

Figure 7.1: Gaussian closure error on the Laplace chain (median and 10–90% band) against the grid error budget measured on identical data ($\rho = 0.85$): the closure error dominates the reference error by 10^4–10^7 at every t, so the measurement is trustworthy where it is used.

3. *Stronger correlation helps, not hurts.* Laplace at $t = 0.05$: error $0.43, 0.27, 0.13$ for $\rho = 0.5, 0.85, 0.95$. Informative neighbours act like extra Gaussian observations and Gaussianise the local tilted posterior; the naive intuition that message errors "propagate farther" at strong coupling is not what happens on chains.

7.4.3 Truncating the matrix versus truncating the inference

The last supervision question — take the exact score and truncate it to tridiagonal: how bad? — admits two inequivalent readings, and keeping them apart turns out to matter. *Truncate the answer* (variant a): band the matrix, $\widehat{S} = -Q_t^{(b)}x$ with entries beyond $|i-j| \le b$ zeroed. *Truncate the algorithm* (variant b): let messages travel at most r hops — exact inference on the window $x_{k-r}, \ldots, x_{k+r}$, the local estimator of Section 5.5.1. Both estimators are linear in x, so their RMS errors over the data law are deterministic traces — no sampling. At equal locality budget ($b = r = 1$, $K = 40$, $\alpha = 0.8$):

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
banded matrix ($b = 1$)	0.210	0.302	0.135	0.0035
truncated inference ($r = 1$)	0.123	0.235	0.130	0.0035

Two lessons. First, the cost of locality is non-monotone in t and peaks at intermediate times — the same mid-diffusion window where the band of Q_t is fullest (Chapter 5) and where generative sampling spends its critical steps. Second, at small t *cutting the algorithm beats cutting the answer* (12.3% vs 21.0% at $t = 0.05$): the windowed inference solves a correctly normalised problem on its window, while zeroing entries of an inverse produces a matrix that is the inverse of nothing and misstates the diagonal. The two coincide at large t, where all off-diagonal content dies. The error of either truncation decays geometrically in the allowed range, at the rate $q(\alpha, t)$ of Theorem 5.11.

7.5 What this chapter established

Exact (proved, and verified to 10^{-12}–10^{-15}): the factor graph of the noisy chain is a tree, so the two BP sweeps compute all smoothing marginals exactly in $O(K)$ message updates; on the Gaussian chain the messages close algebraically in the Gaussian family — no CLT — so the updates cost $O(1)$ each and BP, which is exactly the Kalman filter plus RTS smoother, reproduces the matrix score to machine precision. *Numerical* (measured against validated exact references): the Gaussian closure error on non-Gaussian chains — $\sim$ 20–40% at small t, $< 10^{-3}$ at large t, worst for bimodal innovations, decreasing in ρ — with the score direction much more robust than its magnitude; and the locality comparison, where truncated inference beats the truncated matrix at small t. *Heuristic*: the architecture reading — local, convolution-like score networks with receptive field sized by $q(\alpha, t)$ (or grown with t) are near-optimal outside the mid-diffusion window, and the exactly quantified penalty inside it is the price of locality, not an artefact. The per-node closures of statistical physics, and the exact phase boundary they develop on this very model, are worked out in Appendix C; the findings, limitations, and follow-up programme are consolidated in the next chapter.

Chapter 8

Discussion and Conclusions

8.1 Summary of findings, by research question

RQ1 — the exact joint score and its lifecycle. For the stationary AR(1) chain under coordinatewise OU corruption, the joint score is the linear field $S(x,t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$, equal — by the score–posterior identity, for any prior — to the amplified posterior-mean form $(e^{-t}\mathbb{E}[a|x] - x)/\Delta_t$. Its coupling structure is a complete, closed-form lifecycle: exactly tridiagonal at $t = 0$ (the Hammersley–Clifford fingerprint of Markovianity), band-filling at rate $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$ — with a transcription error of an earlier note corrected and the correction audited — and returning to the identity at the universal rate e^{-2t}. In the bulk, one dimensionless dial $q(\alpha,t) \in (0, |\alpha|]$ governs simultaneously the posterior correlation length and the geometric decay of evidence influence, with the exactly right limits $q \to 0$ ($t \to 0$) and $q \to |\alpha|$ ($t \to \infty$).

RQ2 — local computation of a global object. The posterior of any Markov chain observed through per-frame noise lives on a tree-shaped factor graph, so sum–product message passing computes every marginal — hence every score component — exactly, in $O(K)$ message updates; the total *cost* is linear only when the messages admit a finite parametrisation, which is precisely what the Gaussian case delivers. On the Gaussian chain the messages remain Gaussian by *algebraic closure* (two lemmas: products and affine marginalisation), not by any central-limit argument; the resulting algorithm is, update for update, the Kalman filter plus the Rauch–Tung–Striebel smoother, and it reproduces the matrix score to 10^{-14}. The joint diffusion score of a Markov sequence *is* a smoothing problem — a bridge that imports seventy years of estimation theory into generative modelling.

RQ3 — the price of the Gaussian closure. The answer splits into three cleanly separated statements. On the Gaussian chain, the *score* is closure-independent: mean field, BP, and AMP all solve the same linear system for the posterior mean, so all return the exact score. The *variance* closure is where AMP pays: the per-node (TAP) closure differs from the exact cavity by a single factor of two in the discriminant, which produces an exact breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$ — a sharp, closed-form verdict on the "central limit theorem on two data points" (Appendix C). Beyond Gaussianity, on Laplace and other non-Gaussian chains measured against validated grid quadrature, the Gaussian message closure costs tens of percent of relative score error at small diffusion time (median 0.39 at $t = 0.02$; worst for bimodal innovations), decaying below 10^{-3} once the diffusion noise Gaussianises the posteriors — with the error amplified at small t by the exact factor $e^{-t}/\Delta_t \sim 1/2t$, and mitigated, counterintuitively, by *stronger* chain correlation.

RQ4 — implications for learned scores. Locality has a price and the price has a formula. The influence of evidence at distance d decays as $q(\alpha, t)^d$; the error of a radius-r local estimator decays as q^r; and the penalty peaks in the mid-diffusion window, where the band of Q_t is fullest — precisely where generative sampling spends its critical steps. At equal locality budget, truncating the *inference* (windowed message passing) beats truncating the *matrix* (banding the score): 12.3% against 21.0% relative RMS at $t = 0.05$. For architecture design the readings are direct: local, convolution-like score heads are near-optimal at small and large t; the receptive field should be sized as $r \gtrsim \xi \log(1/\varepsilon)$

with $\xi = 1/\log(1/q)$, or grown with diffusion time; and if a local computation must be used, it should be a local *algorithm*, not a banded linear readout.

8.2 Contributions to the literature

To the exactly solvable programme in the statistical physics of diffusion (Biroli and Mézard, 2023; Biroli et al., 2024; Ghio et al., 2024), this thesis adds the missing axis of *internal structure*: a solvable model in which a single data point has a dynamics of its own, and in which the interplay of internal correlation (α) and diffusion time (t) is charted exactly. To the message-passing literature (Mézard and Montanari, 2009; Zdeborová and Krzakala, 2016), it contributes a minimal, fully worked case where BP and AMP are *both* solvable and *provably different* — the factor-two discriminant, the phase boundary $t_c(\alpha)$, $\alpha_c = \sqrt{2} - 1$ — together with a quantitative audit of the Gaussian closure on chains, the regime where its classical CLT justification fails by construction. To the practice of diffusion modelling, it offers exact baselines: any learned score on Markov-like sequence data can now be compared, regime by regime, against a closed-form ground truth with known locality laws.

8.3 Limitations

Honesty about scope is part of the method. (i) The exact solutions concern *one-dimensional* state per frame and *linear* (AR(1)) dynamics; richer per-frame state keeps the tree structure but complicates the algebra. (ii) The non-Gaussian results are numerical, not closed-form: grid BP is an exact reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3\,\mathrm{d}x$), and the Laplace closure error is a measured curve, not a formula. (iii) The locality laws are proved for the Gaussian bulk; their non-Gaussian counterparts are only structurally delimited (Chapter 6). (iv) No learning is performed: the architecture implications are readings of exact inference results, not training experiments. (v) An earlier conjecture on low-rank structure of the Laplace $K = 2$ Hessian proved regime-dependent and is deliberately not asserted. (vi) Reverse-generation dynamics — how score errors integrate along the sampling trajectory — was explored only preliminarily and is not claimed here.

8.4 Future work

Four directions follow naturally, in increasing ambition. *Mixture messages.* The single-Gaussian closure fails worst on bimodal beliefs; a two-component mixture closure on the same sweeps would separate the message-representation error from the model-mismatch error, and is the clearest next algorithmic step. *Discrete chains.* Replacing the continuous state with a finite alphabet (transition-matrix dynamics, OU noise only in the channel) makes BP messages finite vectors — exact inference with no closure at all — and connects directly to existing tree-reconstruction theory; this was the alternative route sketched in supervision. *Approximate Markovianity.* Real sequences are only approximately Markov; preliminary experiments with a chain-plus-global-latent model (rank-one residual) and with hybrid BP-plus-learned-residual scores suggest the BP baseline absorbs most of the structure at a fraction of the parameters, and deserve a systematic treatment. *Reverse dynamics.* Integrate the reverse SDE under exact, Gaussian-closed, and truncated scores, and measure where along the trajectory the closure and locality errors actually matter for sample quality — the pointwise-versus-dynamical error distinction.

8.5 Closing remark

The thesis began with a black box — a neural network asked to learn $\nabla \log p_t$ — and ended with a dial, $q(\alpha, t)$, and a phase boundary, $\alpha_c = \sqrt{2}-1$. The distance between those two endpoints is the argument for exactly solvable models: when the data have structure, the score inherits it in a computable way, and both the algorithms that exploit it (message passing) and the shortcuts that betray it (closures, truncations) can be judged against ground truth rather than against each other. For sequential data with Markov structure, this programme is now complete at the Gaussian core and quantitatively mapped at its non-Gaussian rim; the frontier — mixtures, discreteness, approximate Markovianity, dynamics — is stated, and it is sharp.

Appendix A

Gaussian Identities Used Throughout

This appendix collects, with proofs, the handful of Gaussian identities on which Chapters 5–7 rely. Notation: a Gaussian density in *moment* parameters is $\mathcal{N}(a; m, v)$; in *natural* parameters it is $\exp(-\frac{\lambda}{2}a^2 + \theta a + \text{const})$ with $\lambda = 1/v$, $\theta = m/v$.

A.1 Quadratic forms and Gaussian normalisation

For $J \succ 0$ (symmetric positive definite) and any h,

$$\int_{\mathbb{R}^K} \exp\Big(-\tfrac{1}{2}a^\top J a + h^\top a\Big)\,\mathrm{d}a = \frac{(2\pi)^{K/2}}{\sqrt{\det J}} \exp\Big(\tfrac{1}{2}h^\top J^{-1} h\Big), \tag{A.1}$$

by completing the square, $-\frac{1}{2}a^\top J a + h^\top a = -\frac{1}{2}(a - J^{-1}h)^\top J (a - J^{-1}h) + \frac{1}{2}h^\top J^{-1} h$, and translating. Consequently a density $p(a) \propto \exp(-\frac{1}{2}a^\top J a + h^\top a)$ is the Gaussian with mean $m = J^{-1}h$ and covariance J^{-1}: *reading off* (J, h) *from a log-density is the fastest route to posterior parameters*, used in Proposition 5.5.

A.2 Products of Gaussians in the same variable

$$\mathcal{N}(a; m_1, v_1)\,\mathcal{N}(a; m_2, v_2) \;\propto\; \mathcal{N}\Big(a;\, m, v\Big), \qquad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}. \quad \text{(A.2)}$$

Proof. Add the exponents; the sum of two quadratics in a is a quadratic in a whose a^2-coefficient is $-\frac{1}{2}(1/v_1 + 1/v_2)$ and whose linear coefficient is $m_1/v_1 + m_2/v_2$; apply (A.1) in one dimension. In natural parameters the rule is addition: $(\lambda, \theta) = (\lambda_1 + \lambda_2, \theta_1 + \theta_2)$ — which is why the BP updates of Chapter 7 are two additions per message. □

A.3 Affine propagation (marginalisation)

If $a' = \alpha a + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2)$ independent of $a \sim \mathcal{N}(m, v)$, then

$$\int \mathcal{N}(a'; \alpha a, \sigma^2)\,\mathcal{N}(a; m, v)\,\mathrm{d}a = \mathcal{N}\Big(a';\, \alpha m,\, \alpha^2 v + \sigma^2\Big). \quad \text{(A.3)}$$

Proof. a' is a sum of independent Gaussians, hence Gaussian; its mean is αm and its variance $\alpha^2 v + \sigma^2$ by linearity of expectation and independence. Read in the reverse direction (solving for a given a Gaussian summary of a'), the same computation gives $\mathcal{N}\Big(a;\, m'/\alpha,\, (v' + \sigma^2)/\alpha^2\Big)$, the backward half-step of Chapter 7. □

A.4 Gaussian conditioning

For a jointly Gaussian pair with precision blocks $\left(\begin{smallmatrix} J_{11} & J_{12} \\ J_{12}^{\top} & J_{22} \end{smallmatrix}\right)$, the conditional of block 1 given block 2 is Gaussian with precision J_{11} (conditioning *reads off* the precision block) and mean shifted by $-J_{11}^{-1} J_{12}\, a_2$; marginalisation, dually, is trivial in covariance parameters. This precision/covariance duality — conditioning is local in J, marginalising is local in Σ — is the algebraic engine behind the cavity recursion (C.1): integrating a neighbour out of a tridiagonal precision produces the rank-one correction $-\beta^2/\lambda$ on the adjacent diagonal entry, by the Schur complement

$$\begin{pmatrix} \lambda & \beta \\ \beta & J_d \end{pmatrix} \longrightarrow J_d - \frac{\beta^2}{\lambda}. \tag{A.4}$$

A.5 Woodbury and the two routes to the score

The Woodbury/Sherman–Morrison–Woodbury identity, $(A+UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$, specialised to $\Sigma_t = \Delta_t I + e^{-2t}\Sigma_0$, connects the noisy precision Q_t and the posterior precision $J = (e^{-2t}/\Delta_t)I + Q_0$:

$$Q_t = \frac{1}{\Delta_t}\left(I - \frac{e^{-2t}}{\Delta_t}\, J^{-1}\right), \tag{A.5}$$

which is exactly the operator form of the score–posterior identity: substituting the posterior mean $m = J^{-1}h = J^{-1}(e^{-t}/\Delta_t)x$ into $S = (e^{-t}m - x)/\Delta_t$ gives $S = -Q_t x$ identically. The machine-precision agreement of the two score routes (Section 5.3) is the numerical shadow of (A.5).

Appendix B

Long Derivations and Supporting Theory

This appendix collects derivations that the body of the thesis uses but that would interrupt its argument if carried out in place. Each entry is self-contained and referenced from the point of use.

B.1 Detailed balance for Metropolis–Hastings

Derivation B.1 (MH kernel satisfies detailed balance). With target π, proposal g, and the acceptance rule (3.6), the effective transition density for an accepted move $x \neq x'$ is $M(x'|x) = g(x'|x)\,A(x \to x')$. Compute the flux from x to x':

$$\pi(x)\,M(x'|x) = \pi(x)\,g(x'|x)\,\min\left\{1, \frac{\pi(x')\,g(x|x')}{\pi(x)\,g(x'|x)}\right\} = \min\left\{\pi(x)\,g(x'|x),\ \pi(x')\,g(x|x')\right\},$$

where the second equality uses $c\min\{1, b/c\} = \min\{c, b\}$ for $c > 0$. The final expression is *symmetric* under $x \leftrightarrow x'$; therefore

$$\pi(x)\,M(x'|x) = \pi(x')\,M(x|x'),$$

which is detailed balance (3.3). The rejection branch (staying at x) contributes only to the diagonal of the kernel and cannot violate the balance of fluxes between distinct states. Stationarity of π follows by integrating the balance over x; with an irreducible, aperiodic proposal, the convergence theorem of Section 3.1 applies. $\square$

Appendix C

AMP, TAP, and the Bulk Fixed Point

This appendix develops the per-node Gaussian closures of statistical physics — AMP, the TAP equations — on the Gaussian chain, and proves the exact breakdown announced in Remark 7.7. The results are complete and audited; they are collected here rather than in Chapter 7 because they sit off the thesis's main line. The punchline: on the chain, AMP still returns the *exact score*, but its variance closure develops an exact breakdown time $t_c(\alpha)$ with critical coupling $\alpha_c = \sqrt{2} - 1$ — the supervision remark that "a central limit theorem on two data points is usually not right", turned into a theorem with a phase diagram.

C.1 The bulk cavity fixed point of BP

Chapter 5 solved the bulk covariance by matrix algebra; the same formulas re-emerge from the message recursion, which is worth exhibiting because it localises *where* BP and AMP differ. In the bulk, parametrise the forward message (evidence absorbed) by its precision λ_k; the closure lemmas of Section 7.3 give the scalar recursion

$$\lambda_{k+1} \;=\; J_d - \frac{\beta^2}{\lambda_k}, \tag{C.1}$$

with J_d, β the bulk parameters (5.19): absorbing one neighbour's summary costs the Gaussian-integration correction $-\beta^2/\lambda$. The fixed points are $\lambda^\pm = \left(J_d \pm \sqrt{J_d^2 - 4\beta^2}\right)/2$, and since $J_d - 2|\beta| = e^{-2t}/\Delta_t + (1-|\alpha|)^2/\sigma_\eta^2 > 0$ (the AM–GM margin of Chapter 5), the attractive upper fixed point

$$\boxed{\lambda^* = \frac{J_d + \sqrt{J_d^2 - 4\beta^2}}{2}} \tag{C.2}$$

exists for *every* $\alpha \in (-1, 1)$ and $t > 0$, with $|f'(\lambda^*)| = \beta^2/\lambda^{*2} < 1$: the sweep converges geometrically from any initialisation. Recombining the two directions and the local term, and using the fixed-point relation $\beta^2/\lambda^* = J_d - \lambda^*$,

$$\Lambda_{\text{bulk}} = J_d - \frac{2\beta^2}{\lambda^*} = 2\lambda^* - J_d = \sqrt{J_d^2 - 4\beta^2} \quad \Longrightarrow \quad V_{\text{exact}} = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \tag{C.3}$$

in exact agreement with Theorem 5.10. *Belief propagation on the Gaussian chain has no phase transition*: the cavity fixed point exists always, and the discriminant $J_d^2 - 4\beta^2$ is the quantity to watch as we now modify the closure.

C.2 AMP/TAP: same score, different variance

AMP — the TAP equations of spin-glass theory (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016) — is the message passing one uses when the factor graph is *dense*: each message is then a sum of many weak contributions, approximately Gaussian by a central-limit argument, and one tracks per-*node* means and variances instead of per-*edge* messages, with the Onsager term correcting the self-influence. Rigorous derivations of AMP from BP exist in exactly that dense limit (Genovese and Piana, 2026). Our chain is the opposite regime in every respect — diagonal measurement, coupling in the prior, degree two — so it is the cleanest possible place to see what the per-node closure costs. Viewing the posterior as a Gaussian MRF with precision J and field h, the three schemes solve, at each node,

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \qquad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k\neq i} J_{ik}^2 V_k}; \qquad \text{BP: } V_i = (J^{-1})_{ii}. \tag{C.4}$$

The AMP closure is the BP cavity with the "exclude the receiving neighbour" correction dropped ($V_{k\to i} \approx V_k$): an $O(1/N)$ change on a dense graph, an $O(1)$ change on a chain with two neighbours.

Theorem C.1 (The score is closure-independent)**.** *For a Gaussian posterior, the fixed-point condition for the* means *is the same linear system $Jm = h$ under mean field, BP, and AMP. All three schemes hence return the exact posterior mean and — via the score–posterior identity — the same, exact, joint score, for every (K, α, t), regardless of whether their variance closures are accurate or even solvable.*

The proof is one line: in each scheme the mean update at node i reads $m_i \leftarrow (h_i - \sum_{k\neq i} J_{ik} m_k)/J_{ii}$ at the fixed point — the closures modify the uncertainty bookkeeping, not the linear relation among means — and the unique solution of $Jm = h$ is the exact posterior mean. Numerically: AMP means agree with exact means to 4.3×10^{-12}, and AMP scores with BP scores to 1.7×10^{-12}, across all tested chain configurations.

For the *variances*, the story splits. In the bulk the AMP closure (C.4) reads $V = 1/(J_d - 2\beta^2 V)$ — compare BP's cavity $V_{\text{cav}} = 1/(J_d - \beta^2 V_{\text{cav}})$: *the entire difference between BP and AMP on the chain is a factor of two*, both neighbours included where BP excludes one.

Theorem C.2 (AMP bulk variance, breakdown time, critical coupling)**.** *The AMP bulk closure is the quadratic $2\beta^2 V^2 - J_d V + 1 = 0$, with physical root*

$$V_{\text{AMP}} = \frac{J_d - \sqrt{J_d^2 - 8\beta^2}}{4\beta^2}, \tag{C.5}$$

which exists iff $J_d \geq 2\sqrt{2}\,|\beta|$ — strictly stronger than BP's always-true $J_d > 2|\beta|$. Since $J_d(t)$ decreases monotonically in t, the existence condition fails, if ever, past a single time;

solving it as an equality (with $u := e^{-2t}$, $e^{-2t}/\Delta_t = u/(1-u)$) gives

$$\boxed{t_c(\alpha) = -\tfrac{1}{2}\log\frac{g}{1+g}, \qquad g(\alpha) = \frac{2\sqrt{2}\,|\alpha| - 1 - \alpha^2}{1-\alpha^2},} \tag{C.6}$$

finite exactly when $g > 0$, i.e.

$$\boxed{|\alpha| \;>\; \alpha_c \;=\; \sqrt{2}-1 \;\approx\; 0.4142.} \tag{C.7}$$

For $|\alpha| \le \alpha_c$ the AMP variance has a fixed point at every diffusion time; for $|\alpha| > \alpha_c$ it ceases to exist for all $t > t_c(\alpha)$. In the weak-coupling regime the two closures agree to second order, with

$$V_{\mathrm{AMP}} - V_{\mathrm{exact}} = \frac{2\beta^4}{J_d^5} + O\!\left(\frac{\beta^6}{J_d^7}\right) \;>\; 0: \tag{C.8}$$

AMP always over*estimates the bulk variance, at fourth order in the coupling.*

Toolbox C.1 — Roots, branch, and the phase boundary

Branch selection. The quadratic has roots $V_\pm = (J_d \pm \sqrt{J_d^2 - 8\beta^2})/(4\beta^2)$; as $\beta \to 0$, expanding the square root gives $V_- \to 1/J_d$ (the decoupled answer) while V_+ diverges — V_- is the physical branch, and the attractive fixed point of the damped iteration used in practice. *Critical coupling.* $g > 0 \iff \alpha^2 - 2\sqrt{2}|\alpha| + 1 < 0 \iff |\alpha| \in (\sqrt{2}-1, \sqrt{2}+1)$; intersecting with $|\alpha| < 1$ leaves $|\alpha| > \sqrt{2}-1$. Sanity check at $t = \infty$: there the posterior is the bare prior, and the existence condition applied to Q_0 itself reads $(1+\alpha^2) \ge 2\sqrt{2}|\alpha|$ — AMP applied to a bare AR(1) chain works iff $|\alpha| \le \sqrt{2}-1$, which is (C.7) read backwards. *Weak coupling.* With $\epsilon := \beta^2/J_d^2$: $(1-4\epsilon)^{-1/2} = 1+2\epsilon+6\epsilon^2+O(\epsilon^3)$ for the exact variance; $1-\sqrt{1-8\epsilon} = 4\epsilon+8\epsilon^2+O(\epsilon^3)$, divided by $4\epsilon J_d$, gives $V_{\mathrm{AMP}} = (1+2\epsilon+8\epsilon^2+O(\epsilon^3))/J_d$; subtract. □

Numerically, on a $K = 12$ chain at $\alpha = 0.8$: the AMP score error stays at 10^{-13}–10^{-12} for every t, while the variance error is 1.3×10^{-4} at $t = 0.05$, 3.5×10^{-2} at $t = 0.2$, and past $t \approx 0.23$ the variance update has no positive fixed point at all. The audit further verifies: the closed form (C.5) against the converged iteration to 2.5×10^{-12}; the

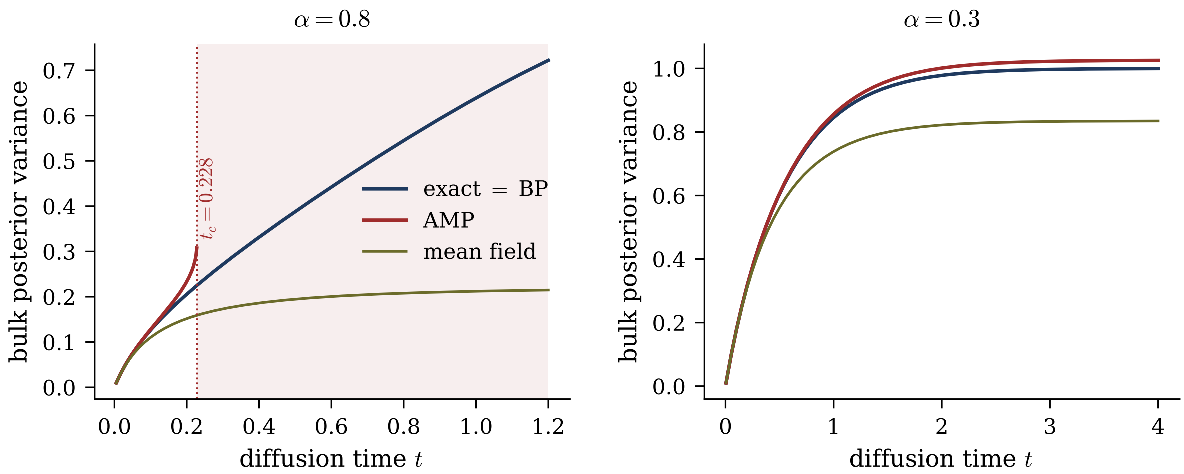

Figure C.1: The three closures in closed form. Left ($\alpha = 0.8 > \alpha_c$): exact = BP, AMP (C.5), and mean field; the AMP branch terminates with a square-root singularity at the exact breakdown time $t_c(0.8) = 0.228$ from (C.6) (shaded: no fixed point). Right ($\alpha = 0.3 < \alpha_c$): AMP exists at every t and slightly overestimates the variance, as predicted by (C.8).

existence condition at all 180 points of an (α, t) scan; t_c against the bisected flip point of the iteration to 6.6×10^{-6} relative; convergence below α_c even at $t = 50$; and the weak-coupling ratio $(V_{\text{AMP}} - V_{\text{exact}})/(2\beta^4/J_d^5) \to 1$ within 0.4%.

The supervision remark that framed this appendix — the Gaussian message assumption is a CLT statement, and "a central limit theorem on two data points is usually not right" — is thus made exact, with both halves landing precisely where predicted. The *conjecture* (Gaussian message passing on the solved case recovers the known score) holds exactly: for BP by algebraic closure, for AMP because the mean fixed point is closure-independent. The *objection* targets the variance closure, and there it is not merely right but quantitative: a factor two in a discriminant, a phase boundary (C.6), and a critical coupling $\sqrt{2} - 1$. AMP is not wrong — on a dense benchmark (random $N = 600$ SPD precision) the same closure gets variances to 1.7×10^{-3} where mean field errs at 4×10^{-2} — it is built for a different graph.

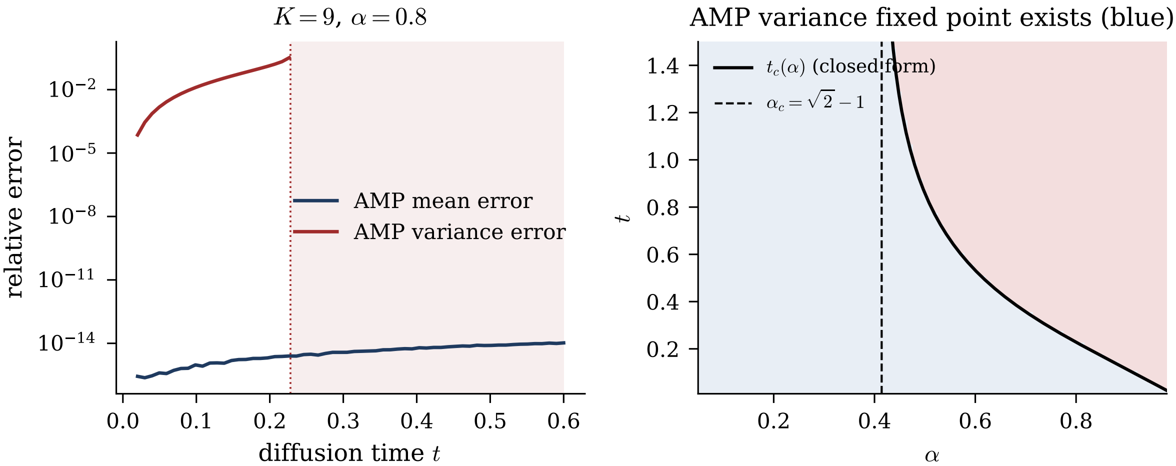

Figure C.2: BP vs AMP on a finite chain ($K = 9$, $\alpha = 0.8$). Left: the AMP *mean* error ($\sim 10^{-12}$, flat in t: the score is closure-independent, Theorem C.1) against the AMP *variance* error, which grows with t until the closure stops converging (shaded). Right: existence of the AMP variance fixed point in the (α, t) plane; the black curve is the exact boundary $t_c(\alpha)$ of (C.6), the dashed line $\alpha_c = \sqrt{2} - 1$; theory and iteration agree at all 180 scanned points.

Appendix D

Reproducibility

D.1 Repository layout

All source material of this thesis lives in a single repository, organised as follows (the thesis directory is self-contained: the listed code and every figure file are committed alongside the LaTeX source).

Path	Content
`thesis/`	this document: LaTeX source, figures and figure script
`research/gaussian-bp/`	Gaussian chain note, code, notebooks, 72-check audit
`research/bp-from-scratch/`	independent BP derivation and implementation
`research/unified-note/`	consolidated working note, code, 55-check audit
`research/nongaussian-bp/`	grid BP vs Gaussian-projected BP experiments
`research/gaussian-ar1-bp/`	Gaussian AR(1) package (BP = matrix score)
`research/bp-generalization/`	general non-Gaussian BP formulation notes
`research/session-summaries/`	source working notes of the project
`research/initial-experiments/`	early diffusion-setup experiments and figures

D.2 Audit protocol

Every closed-form claim is verified by a deterministic check against an independently computed reference (brute-force linear algebra, dense grid quadrature, or finite differences), collected in audit drivers that exit non-zero on any failure:

- `research/gaussian-bp/code/numerical_audit.py` — 72 checks covering: Σ_t assembly and inversion; score–posterior route vs matrix score; the $K = 2$ closed forms; Convention-A BP vs matrix posterior (means, variances, scores); bulk variance, covariance decay q^d, and the recombination identity; AMP mean/score closure-independence; the AMP variance formula, existence boundary, breakdown time, and weak-coupling law; the locality slopes; the band-fill coefficients (factorial version rejected); spectral preservation and large-t return.
- `research/unified-note/code/numerical_audit.py` — 55 checks covering the consolidated note, including the Laplace $K = 1$ quadrature checks (score–posterior route vs finite differences; characteristic function vs direct integration) and the one-step boundary-message identity.
- `research/nongaussian-bp/` — experiment drivers `exp_01`–`exp_05` with logged master-error-identity residuals (10^{-15}–10^{-12} throughout), grid self-convergence gates, and per-trial CSV outputs; the grid calibration of Section 7.4 is experiment 01.

All checks were re-executed on the build machine at the time of writing: 72/72 and 55/55 pass on the audit drivers, and all experiment gates are green.

D.3 Figure-to-code map

Every figure of this thesis is regenerated by a single script, `figures/make_figures.py`, which recomputes each panel from the closed forms (or quadrature) of the corresponding chapter; the only figure that reads stored experimental data is Figure 7.1, built from the per-trial CSV of `research/nongaussian-bp` experiment 02. All figures are vector PDF, white background.

Figure	File	Source of the numbers
5.1	`fig_spectral.pdf`	closed forms, Ch. 5
5.2	`fig_precision_lifecycle.pdf`	closed forms, Ch. 5
5.3	`fig_band_fill.pdf`	closed forms, Ch. 5
5.4	`fig_tridiag_loss.pdf`	closed forms, Ch. 5
5.5	`fig_local_vs_full.pdf`	closed forms, Ch. 5
6.1	`fig06_K1_density.pdf`	quadrature, Ch. 6
6.2	`fig07_K1_score.pdf`	quadrature, Ch. 6
6.3	`fig08_K1_score_limits.pdf`	quadrature, Ch. 6
7.1	`fig_laplace_closure.pdf`	`nongaussian-bp` exp. 02 CSV
C.1	`fig_bulk_variance.pdf`	closed forms, App. C
C.2	`fig_bp_vs_amp.pdf`	closed forms + iteration, App. C

D.4 Environment

Python $\geq$ 3.10 with NumPy, SciPy, and Matplotlib; no GPU, no neural network framework, no stochastic training. All experiments are deterministic given the seeds recorded in the experiment configurations; the error measures of Sections 5.5.1 and 7.4.3 are seed-free by construction (closed-form traces). This document compiles with a single run of `tectonic main.tex`.

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